



MSc Robotics

B31VZ - MSc Project - Dissertation

**RAPTOR-4-AUV: Generalized 6-DOF Underwater
Control via GPU-Parallel Meta-Imitation Learning
with GRU**

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August 17, 2026

Year of Graduation: 2026

Word count: 14900

Dissertation submitted in partial fulfillment of the requirements
for the degree of MSc at Heriot-Watt University

Abstract

Controllers for autonomous underwater vehicles are usually tuned for one vehicle and need retuning whenever hull, payload or ballast changes. Recent aerial work shows that a single recurrent policy, distilled from a population of per-vehicle reinforcement-learning teachers, can control many physically distinct vehicles without retuning. This dissertation transfers that approach to 6-degree-of-freedom underwater vehicles. It asks whether one recurrent policy can match per-vehicle specialists across a 5–150 kg bank and exceed them off-nominal, by what mechanism it adapts, and how it compares against a matched domain-randomisation baseline. We trained one Proximal Policy Optimisation teacher for each of 1,000 sampled vehicles in a GPU-parallel simulator, and distilled them into a single gated-recurrent-unit student by meta-imitation. The student matched the specialists on nominal physics and outperformed them under isolated stressors, reducing position error under strong buoyancy from 19.2 to 3.5 cm. A linear probe recovered buoyancy from the student’s hidden state on 300 held-out vehicles at $R^2 = 0.87$ while mass, thrust, and shuffle controls decoded at 0.04 or below, evidencing identification of the control-relevant parameter specifically. This required holding each teacher’s buoyancy fixed during distillation, so imitation targets stay self-consistent. To illustrate the pipeline’s worth, buoyancy domain randomised baselines were added and these showed much more promising than anticipated. The single-axis baseline matched or exceeded most distilled students on buoyancy, mid-episode changes and unseen vehicles, and matching the centre-of-gravity axis closed roughly 90% of the attitude gap previously credited to the architecture. A memoryless policy does pay for each added axis globally: its nominal hover error nearly tripling where the same axis cost the recurrent policy almost nothing. Yet, the baselines proved more reliable outside the training distribution, holding full success where students fell to 53–71%. It is concluded that recurrence buys the composition of several identity axes rather than better single-axis adaptation, sacrificing graceful degradation beyond its training range.

Keywords: autonomous underwater vehicles; reinforcement learning; teacher–student distillation; meta-imitation learning; implicit system identification; domain randomisation; buoyancy; recurrent policies; sim-to-sim transfer; baseline validation.

Acknowledgements

I hereby declare that this report is entirely my own work and that all sources have been properly acknowledged.

I would like to express my sincere gratitude to everyone who supported me throughout the completion of this MSc dissertation.

First, I want to thank my supervisor, Dr. Ignacio Carlucho, for his essential guidance, patience, and support throughout this research project.

I am deeply grateful to the PhD student, Sümer Tunçay, who also guided and supported me and whose work provided the foundation upon which this project was built.

My thanks also go to my friend and colleague, Thomas Bonnier, for sharing this journey under the same supervision and for his helpful input and support along the way.

I extend my appreciation to the authors of the RAPTOR paper, whose work served as the core basis for my research, as well as the Ocean Systems Lab (OSL) for granting me access to their servers and facilities.

Finally, I owe a special thank you to my parents for giving me the opportunity to pursue my Master's degree at Heriot-Watt University. Their constant support and encouragement made this achievement possible.

Signed: Arne Jacobs

Date: August 17, 2026

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1 Introduction

1.1 Background and Motivation

Autonomous underwater vehicles (AUVs) and remotely operated vehicles (ROVs) are now central to seabed survey and marine geoscience [1], offshore-infrastructure inspection and maintenance [2, 3] and sustained ocean observation [4]. Despite this breadth, the controllers that stabilise these vehicles remain vehicle-specific. A low-level controller is identified and tuned for one hull, mass, thruster layout and ballast configuration, and change to any of these requires retuning. Adding a payload, swapping a sensor or fitting a manipulator can therefore degrade control until an engineer re-identifies the dynamics and re-tunes the gains, consuming expert time, ship time and money. The classical marine control literature is mature and effective, but its guarantees are dependant on an accurate vehicle-specific model [5], and that reliance is exactly what makes the approach costly to scale across a fleet of heterogeneous platforms.

What makes the underwater case distinctive is not the number of degrees of freedom but the structure of the dominant off-nominal force. Net buoyancy is set by the hull's displaced volume and, for an imperfectly ballasted vehicle, appears as a slowly varying force bias. More importantly, it acts at the centre of buoyancy, so any offset from the centre of gravity produces a persistent restoring moment coupling directly into roll and pitch [6]. These terms are properties of the vehicle's configuration and payload rather than transient disturbances to be rejected [7]. A policy that has learned to hold depth and attitude on one vehicle will therefore mis-trim on a differently ballasted one. This weakness is only partially analogous to mass and thrust-to-weight variation in quadrotors, where the gravity is fixed in magnitude and direction and acts at the centre of mass.

Two developments make a population-scale response feasible. The first is GPU-parallel rigid-body simulation. Frameworks built on JAX and MuJoCo-XLA execute thousands of physics environments in parallel on a single GPU [8, 9]. This makes training one policy per vehicle across a large bank becomes a matter of days rather than weeks. In this work a single teacher trained in approximately fifteen minutes and eight teachers were trained concurrently on one GPU (NVIDIA A40, 46 GB or RTX 5090). The second is the demonstration, in the aerial domain, that a single recurrent policy distilled from many specialist teachers can control a whole population of different drones without retuning [10].

1.2 Problem Statement and Origin of the Study

Teacher–student distillation into a single adaptive policy has been demonstrated for aerial vehicles [10], and fast single-vehicle reinforcement learning has been demonstrated for underwater vehicles [11]. No prior work combines both to distil a population of 6-DOF underwater vehicles into one policy, in the regime where hydrostatics, and buoyancy in particular, dominate the off-nominal behaviour.

My colleagues’ earlier attempts to handle buoyancy variation by randomisation alone had not produced a satisfactory controller, and that experience motivated the RAPTOR-style population pipeline on the group’s existing fast 6-DOF AUV training stack. The working assumption that framed this study’s design was that a recurrent policy able to infer buoyancy would succeed where a memoryless randomised one had not.

The specific question is whether a single recurrent controller can generalise, without retuning across a bank spanning roughly a thirty-fold mass range (5–150 kg) and correspondingly wide ranges of size, inertia, thrust authority and buoyancy, matching the per-vehicle specialists it was trained from and exceeding them under conditions no specialist can adapt to. A second question is by what mechanism such a single policy adapts: whether it merely reacts to instantaneous error, or whether recurrence lets it maintain an internal estimate of the vehicle it controls. In other words, does it perform a form of implicit system identification? A third, added late and ultimately decisive, is how it compares against the very baseline the project had assumed unsatisfactory.

1.3 Aim and Objectives

The aim of this dissertation is to answer those three questions: whether one recurrent policy can control the bank without retuning, by what mechanism it adapts, and what the architecture buys against a far cheaper baseline. The work is additionally intended to yield a contribution of publishable standard, and Chapter 6 identifies which of its findings are strongest in that respect. The aim is pursued through five objectives:

- O1** Build a parameterised 6-DOF AUV simulation environment and a sampled bank of vehicle hardware configurations covering wide ranges of mass, size, hull shape and buoyancy.
- O2** Train one PPO specialist (“teacher”) for each configuration in that bank.
- O3** Distil the teachers into a single recurrent (GRU) generalist (“student”) by meta-imitation, measure its adaptation to off-nominal buoyancy, added payload and external disturbance. Then probe its recurrent hidden state to establish whether that adaptation constitutes implicit system identification.

- O4** Test transfer of the resulting policy to vehicles outside the training bank, to an independent physics engine (Stonefish) and a physically distinct, bottom-heavy vehicle (BlueROV2-Heavy), and identify the physical gaps that govern where that transfer holds and where it breaks.
- O5** Evaluate the distilled policies against domain-randomisation baselines under the identical evaluation protocol, and determine which of the pipeline’s claimed advantages survive a comparison matched axis by axis.

1.4 Structure of the Dissertation

Chapter 2 reviews marine vehicle dynamics and control, deep reinforcement learning for continuous control, sim-to-real transfer and domain randomisation, teacher–student distillation, GPU-parallel simulation and closes by stating the research gap and the four research questions following from it. Chapter 3 justifies each methodological decision: environment, hardware bank, teacher and student training, the domain-randomisation design distinguishing vehicle-identity parameters from per-episode noise, the baseline control condition, and the evaluation instruments. Chapter 4 presents the results chronologically, from the teacher population through the distillation failure and its diagnosis to the adaptation and transfer results, ending with the late baseline comparison that revises the study’s starting assumption. Chapter 5 interprets those findings and answers the research questions, including what the pipeline is and is not shown to be worth. Chapter 6 concludes with the contributions, implications and limitations, and Chapter 7 sets out recommendations for practice and further research.

2 Literature Review

This chapter brings together the five bodies of literature that jointly define the problem addressed in this dissertation: the dynamics and classical control of underwater vehicles, deep reinforcement learning for continuous control, sim-to-real transfer and domain randomisation, teacher–student distillation for rapid adaptation, and GPU-parallel physics simulation. These research areas are individually mature but have not been combined in the way this work requires, and the chapter is organised to expose precisely where they fail to meet. It begins by stating the review methodology, works through each field critically, and closes by naming the research gap and deriving the aim, objectives and research questions from it.

2.1 Review Methodology

The literature was searched from February to July 2026 across IEEE Xplore, Scopus, ScienceDirect, arXiv, Google Scholar and the OSL discord group where my supervisor directly shared relevant work with the team. Searches combined vehicle terms (autonomous underwater vehicle, AUV, ROV) with method terms (reinforcement learning, domain randomization, sim-to-real, teacher–student, distillation, recurrent policy) and physics terms (buoyancy, hydrostatic).

Recent work was bounded to 2015–2026 while any date was admitted for important contributions. Papers had to concern low-level or dynamic control rather than purely path planning or perception. Pure planning and SLAM studies, surface-only vessels where 6-DOF coupling does not apply, and studies without experimental or simulation validation were excluded. Major and current work were separated deliberately: the major set anchors the theoretical framework (Fossen for marine dynamics [5]; Sutton and Barto for reinforcement learning [12]; Tobin et al. for domain randomisation [13]; Ross et al. for imitation learning [14]; Kumar et al. for privileged adaptation [15]), while the current set establishes the state of the art the contribution is positioned against (RAPTOR [10]; fast AUV learning [11]; Stonefish [16]; MuJoCo Playground [8]; Chaffre et al. [17]).

2.2 Dynamics and Control of Underwater Vehicles

The dynamics of a marine craft are described in the standard formulation by Fossen [5] as a set of 6-DOF rigid-body equations of motion augmented by hydrodynamic effects. The model takes the standard rigid-body-plus-hydrodynamics form

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}, \quad \dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta})\mathbf{v}, \quad (1)$$

where $\mathbf{v} = [u, v, w, p, q, r]^\top \in \mathbb{R}^6$ is the body-fixed linear and angular velocity, $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^\top \in \mathbb{R}^6$ is the pose in the earth-fixed frame, and $\mathbf{J}(\boldsymbol{\eta})$ is the Euler-angle rotation matrix relating the two. $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ is the rigid-body-plus-added-mass inertia matrix, $\mathbf{C}(\mathbf{v}) = \mathbf{C}_{RB}(\mathbf{v}) + \mathbf{C}_A(\mathbf{v})$ the associated Coriolis–centripetal matrix, $\mathbf{D}(\mathbf{v})$ the hydrodynamic damping matrix, $\mathbf{g}(\boldsymbol{\eta})$ the vector of restoring forces and moments due to gravity and buoyancy, and $\boldsymbol{\tau} \in \mathbb{R}^6$ the control wrench delivered by the thrusters. Three of these effects are fundamental for the present work. The first is *added mass* ($\mathbf{M}_A$): as the vehicle accelerates it must accelerate the surrounding fluid, which appears as an inertia-like term that couples translational and rotational motion. The second is *hydrodynamic damping* ($\mathbf{D}(\mathbf{v})$), dominated at operational speeds by a quadratic drag term. The third, and the one that motivates this dissertation, is the *hydrostatic restoring force* ($\mathbf{g}(\boldsymbol{\eta})$). This is the one generated by gravity acting at the centre of gravity (CoG) and buoyancy acting at the centre of buoyancy (CoB). When CoB and CoG do not coincide, their offset produces a restoring moment that acts like a pendulum, righting the vehicle toward a preferred attitude. When the buoyant force does not exactly balance the weight, it leaves a persistent vertical force that the actuators must continuously oppose.

This last point is central. The vertical force imbalance produced by imperfect ballasting is not a disturbance in the usual sense; but a fixed property of a given vehicle in a given configuration. This imbalance stays present for that vehicle’s whole operating lifetime until it is physically re-ballasted. This CoB–CoG offset is what reappears in Chapter 5. It produces the pendulum-like righting moment that explains the attitude behaviour seen in the sim-to-sim transfer: a bottom-heavy vehicle self-rights passively, but resists being commanded into an arbitrary attitude.

The added-mass and coupling terms distinguish the underwater regime from the aerial one. In water, the fluid a vehicle displaces as it accelerates is of comparable density to the vehicle itself. This makes the added-mass contribution a substantial fraction of the rigid-body inertia rather than a negligible correction. It also couples degrees of freedom, so that a commanded force along one axis induces motion and moments about others. These structures vary with geometry in ways mass alone does not capture, which is why a policy learned on one hull cannot be assumed to transfer to another of different shape, and why hull geometry, not only mass, is sampled across the training bank.

A related body of work concerns thruster modelling and allocation. A 6-DOF wrench has to be realised by a fixed set of thrusters whose geometry decides which force-and-moment combinations are even possible. Turning a desired wrench into individual thruster commands is a well-studied problem, solved by a pseudo-inverse of the thruster-configuration matrix or, on a specific platform such as the BlueROV2, by a fixed

mixing matrix, with redundant layouts allowing fault-tolerant reallocation [18]. This step matters directly for Chapter 4 since the sim-to-sim work relies on getting it right. A mistake as small as a sign error or a coordinate-frame mismatch would produce a transfer failure that could easily be mistaken for a policy failure.

Chapter 3's vehicle bank rests on the same hydrodynamic modelling described above. Prestero's identification of the REMUS hull [19] remains the standard worked example of deriving a full 6-DOF hydrodynamic parameter set for a real AUV and validating it against tank data; the mass-scaling relations used here (length as $\text{mass}^{1/3}$, inertia as $\text{mass}^{5/3}$) are the dimensional-analysis skeleton of that kind of identification, applied across a population rather than to a single hull. The mixing matrix used in the sim-to-sim bridge likewise follows the standard treatment of allocation for over-actuated vehicles [20].

Against this dynamical backdrop, classical control of underwater vehicles is well established, and recent reviews trace its development towards learning-based methods [21, 22, 23]. PID control remains the workhorse of practical ROV station-keeping for its simplicity and robustness, with adaptive variants such as self-tuning fuzzy-PID extending it to the coupled heading-and-depth problem [24]. Sliding-mode control addresses bounded but unknown model error, which is precisely the problem this dissertation attacks, and has the longest pedigree in this domain. Yoerger and Slotine [25] first showed that a sliding surface makes underwater trajectory tracking robust to large uncertainty in the hydrodynamic coefficients, and Healey and Lienard [26] extended it to the coupled diving-and-steering problem in the multivariable form still used as a reference design; the cost is actuator chattering, and later work develops the underactuated case [27, 28]. Adaptive control attacks the same uncertainty by estimating the unknown parameters *online*, and is the closest classical ancestor of the implicit identification claimed in this work: Antonelli et al. [29] demonstrate it experimentally on the ODIN vehicle, adapting to payload and hydrodynamic variation without a prior model. Smallwood and Whitcomb [30] address the station-keeping task directly and quantify how much a model-based positioning controller gains over a fixed-gain one on a real vehicle.

Two further families complete the picture. Backstepping exploits the cascaded structure of vehicle dynamics, kinematics driven by dynamics, to construct a stabilising law and its stability certificate together [31, 32]; model-predictive control instead optimises over a receding horizon, handling actuator and state constraints explicitly at the cost of solving an optimisation online [33]. Their strengths must be acknowledged in any critical evaluation: interpretability, stability guarantees under stated assumptions, low online cost, and decades of field deployment. A learned controller that cannot at least match their nominal performance would not be worth deploying.

One critical limitation is common to all of the classical controllers and is the one this dissertation targets. Every classical method requires a vehicle-specific dynamic model to be identified, its parameters re-identified and the controller retuned whenever the vehicle changes. Adaptive and robust control soften this dependence but do not remove it. Adaptive schemes estimate uncertain parameters online but still assume a fixed model structure and bounded parameter variation, and robust schemes tolerate bounded model error but trade performance for a guaranteed envelope. Moreover, both, adapt within a single deployment, against a parameter range the designer fixed in advance. The approach taken here instead learns the adaptation once, across a population of a thousand vehicles, so that no online estimator and no hand-chosen bound is needed at deployment. Carlucho et al. [34] make precisely this argument in motivating learned control, demonstrating an actor–critic controller for AUV low-level thruster control that adapts from raw sensory input and was validated on a real vehicle. This dissertation adopts their framing: that per-vehicle model identification is the principal obstacle to scalable AUV control. It is on that premise that this review turns to reinforcement learning.

2.3 Deep Reinforcement Learning for Continuous Control

Reinforcement learning formalises control as a Markov decision process in which an agent selects actions to maximise expected cumulative reward, policy-gradient methods optimising a parameterised policy by ascending the gradient of that objective [12]. The field’s modern revival began with value-based deep learning [35] before continuous control motivated the actor–critic methods used here, whose two dominant algorithms illustrate a fundamental trade-off. Proximal Policy Optimisation [36] is on-policy, constraining each update with a clipped surrogate objective to prevent destructively large policy changes, yielding stable training at the cost of sample efficiency since each batch is used for only a few updates. Soft Actor–Critic [37] is off-policy, reusing experience from a replay buffer and adding an entropy term, achieving far higher sample efficiency at the cost of hyperparameter sensitivity; the related twin-delayed deterministic policy gradient addresses the value-overestimation plaguing earlier off-policy actor–critics [38].

Table 1: Why the massively-parallel setting inverts the usual PPO-versus-SAC trade-off. The decisive row is the last: with thousands of environments stepped per update, samples are nearly free and wall-clock throughput becomes the binding constraint.

	PPO (on-policy)	SAC (off-policy)
Data reuse	few updates, then discarded	replay buffer, reused many times
Sample efficiency	low	high
Update stability	high (clipped surrogate)	moderate; entropy term to tune
Hyperparam. sensitivity	low	high
Parallel scaling	large synchronous batches	buffer becomes a bottleneck
Binding constraint here	wall-clock	sample count

The choice between them is dictated by the setting, and the axes that decide it are summarised in Table 1. Where environment interaction is expensive, as on real hardware, SAC’s sample efficiency is decisive, but where simulation is cheap and massively parallel the calculus inverts. The bottleneck becomes wall-clock throughput rather than sample count, and PPO’s stability and natural fit to large synchronous batches make it appropriate. As we have thousands of environments stepped in parallel on a GPU the logical choice was the go with PPO to train the teachers.

The recent encouraging results when applying reinforcement learning to AUVs have made the area mature, enough so to attract dedicated reviews [39, 40]. Carlucho et al. [34] demonstrated adaptive low-level control on a real vehicle and more recent work pushes into robustness, for example an adversarial scheme hardening depth tracking by pitting the controller against a learned adversary [41]. Energy-aware formulations of the 6-DOF position task [42] illustrate a contrasting objective, trading tracking accuracy against power where the present work trades it against generality across vehicles. Most directly, Tunçay et al. [11] demonstrate GPU-accelerated learning for 6-DOF AUV position control, training a single-vehicle policy in under two minutes with zero-shot sim-to-real transfer. That work is the immediate technical antecedent of this dissertation, which generalises its single-vehicle method to a population.

The limitations are equally well documented: sample inefficiency, acute sensitivity to reward shaping, absence of stability guarantees, and brittleness when the deployment distribution departs from the training one. Parallel simulation mitigates the first by trading sample count for wall-clock throughput. Reward-shaping sensitivity is a recurring difficulty: the action-regularisation terms used here are a design choice whose weighting readily yields policies that track well but chatter or saturate. The absence of formal stability guarantees is the most consequential limitation for safety-critical marine deployment. Curriculum learning [43] and meta-learned optimisers [44] are alternatives to the large-batch strategy adopted here. Brittleness is precisely the weakness randomi-

sation and distillation are designed to mitigate, and the subject of the next two sections. The literature also predicts on-policy sensitivity to batch size, which Chapter 4 confirms.

2.4 Sim-to-Real Transfer and Domain Randomisation

A policy trained in simulation and deployed in reality confronts the *reality gap*: the systematic mismatch between simulated and real dynamics. Domain randomisation is the dominant response. Tobin et al. [13], in the main statement, argue that if the simulator is randomised widely enough during training “the real world may appear to the model as just another variation”; the policy trains not on one simulator but on a distribution of them, learning behaviour robust to the randomised axes. It is now standard practice to treat the randomised quantities (masses, friction, sensor noise, latencies, appearance) as variation to be tolerated [45].

In marine robotics, randomisation and sim-to-real transfer have been growing rapidly. Chaffre et al. [17] transfer adaptive control parameters for AUV stabilisation under current disturbance, the closest published comparison for the current-rejection experiments of Chapter 4. Lu et al. [46] use *data-informed* randomisation foreshadowing the argument here that not all randomised quantities should be treated identically. Indeed, they shape the distribution with real-vehicle data rather than a naive uniform range. Tunçay et al. [11] achieve zero-shot AUV sim-to-real transfer for position control, and EasyUUV [47] addresses UUV attitude control, speaking to the attitude-transfer difficulty diagnosed in Chapter 5. Validating a policy on a second, independent simulator before committing to hardware occupies a useful middle ground, far cheaper than a hardware campaign yet materially stronger than a single-engine result. The sim-to-sim value depends entirely on the second engine’s independence since transferring between simulators sharing a backend, integrator and hydrodynamics model demonstrates little. However, transferring to an engine differing in all three, and to a vehicle whose mass distribution and hull shape differ from anything in the training bank, probes numerical and physical robustness at once. The strength of such a claim is not that transfer occurred but how far the target differed from the source, and Chapter 3 is explicit about that difference.

A question the literature leaves open is *how far domain randomisation alone can be pushed* before a more elaborate architecture becomes necessary. The received argument for memory is that a memoryless policy randomised over a parameter can only learn the compromise least bad across the whole distribution, whereas a history-based policy can identify which member it faces and behave optimally for it [15]. The argument is sound in principle, but how much it matters in practice is an empirical question. A well-randomised feedforward policy may lose very little in two cases: either the

optimal action barely differs across the distribution of vehicles, or a wrong guess about a parameter produces an error obvious enough for the policy to correct reactively, without ever needing to identify the parameter itself. OpenAI’s dexterous-manipulation work [48] combines aggressive randomisation *with* recurrence and reports memory to help, without isolating the two. Adaptation papers more generally compare against baselines that are weaker (trained for less time, on narrower ranges, or evaluated only where the adaptive method is expected to win). This represents an asymmetry of effort Muratore et al. [49] note as a recurring weakness. The marginal value of recurrence over strong randomisation is thus asserted more often than measured, and this study, having assumed it at the outset, ends by measuring it.

The critical gap in this literature, and the original intellectual core of this dissertation, concerns *how* randomised quantities are treated during training. The domain-randomisation literature almost universally treats them as nuisance noise: a fresh value is sampled every episode, on the assumption that the policy should be invariant to the axis in question. But some parameters are not noise. They are vehicle *identity*: fixed for the operating lifetime of a particular platform, and precisely the thing an adaptive policy is supposed to *infer* and exploit rather than ignore. Mass is borderline; buoyancy, for an underwater vehicle, is unambiguously an identity parameter, with no aerial analogue. Critically, during teacher–student distillation the two categories demand *opposite* handling: a per-episode nuisance parameter should be re-randomised so the student sees the variation the teacher trained under, but an identity parameter must be held fixed at the value the specific teacher being imitated was trained under, or the student is trained on labels that answer the wrong question. No prior work in the distillation or domain-randomisation literature draws this distinction, and Chapter 4 shows that getting it wrong carries a large, measurable cost.

2.5 Teacher–Student Distillation and Rapid Adaptation

Imitation learning trains a policy to reproduce expert actions, but a naive behavioural-cloning student trained only on states the expert visits fails under distribution shift. Indeed, its own small errors compound, carrying it into states the expert never demonstrated. Ross et al. [14], in the DAgger algorithm, resolve this by iteratively training the student on the states *it* visits while continuing to query the teacher for labels there. This is the theoretical justification for the warm-up schedule used in this work’s distillation (Chapter 3), in which the student first imitates on teacher-driven rollouts and then progressively acts on its own.

Privileged learning specialises this for sim-to-real adaptation. Kumar et al.’s Rapid Motor Adaptation [15] trains a base policy with an adaptation module: a privileged

teacher accesses the true environment parameters during training while a student learns to estimate them from a proprioceptive history, and at deployment the student’s estimate drives the base policy. Deployed on a legged robot across sand, mud and grass with no fine-tuning, it is the clearest precedent for the mechanism this dissertation relies on, that is to say that a policy with access to its own response history can perform *implicit system identification*, inferring the physical parameters of the body it controls from how that body responds.

The mechanism RMA [15] formalises where we condition behaviour on an observation history so the policy adapts to unmodelled dynamics recurs under several names. Recurrent policies maintain an internal state summarising that history, and the same effect is pursued in meta-reinforcement learning as in-context or fast adaptation, where an agent trained across a distribution of tasks adapts within a single episode without weight updates [50]. The common thread, explicit in the RL^2 formulation, is that the recurrent state acts as an online estimate of the current dynamics which the policy exploits. What varies is the strength of evidence that such adaptation is genuine inference rather than memorisation of the training distribution. It is on that evidential question, rather than the by-now-standard architecture, that a contribution can most usefully be made.

RAPTOR [10] is the direct antecedent of this work and must be described accurately. It applies meta-imitation learning: one thousand teachers, each trained for a different quadrotor configuration, distilled into a single recurrent student of only 2,084 parameters, which flies ten physically different real quadrotors zero-shot (32 g to 2.4 kg, thrust-to-weight from 1.75 to 12, differing motors, frames, propellers and flight controllers) with no per-vehicle tuning. Two features of its formulation are primordial for Chapter 5 and were confirmed against the paper’s own text. First, RAPTOR is a Bayes-adaptive partially observable MDP in which vehicle parameters are sampled *once at the start of each episode and held constant*, for both teacher training and distillation: fixed identity, not per-step noise. Second, the celebrated mid-flight weight-drop demonstration is a *zero-shot generalisation* result, not an explicitly trained behaviour. These respectively support the identity-parameter argument and the zero-shot-payload result reported later.

Structurally, RAPTOR’s pipeline has three stages, and the present work adopts all three while changing what is randomised at each. A *population* stage, in which one specialist is trained per sampled vehicle. A *distillation* stage, in which those specialists supply action labels to a single recurrent network under a DAgger-style schedule. And a *deployment* stage, in which the recurrent state alone carries whatever the policy has inferred about the vehicle. Figure 2 in Chapter 3 shows the version built here. Two differences are worth stating early. First, the specialists here are underwater rather than aerial, so the identity parameter they span is buoyancy rather than thrust-to-weight. Secondly, this work adds a

matched single-policy randomisation baseline running through the identical evaluation. This is a fourth element RAPTOR does not have and what makes the architectural claim testable rather than assumed.

Using recurrence the drone adapts to a sudden added mass. However, how can we tell that the policy really *infers* the vehicle’s dynamics rather than memorising a lookup over the training set? RAPTOR provides the strongest evidence available in this literature: a linear probe of the student’s latent state recovers thrust-to-weight at $R^2 = 0.949$, showing the identity parameter is explicitly and linearly encoded. The methodology is inexpensive to replicate and directly addresses the memorisation-versus-inference question for any recurrent controller; Chapter 4 applies it here with buoyancy as the target.

2.6 GPU-Parallel Physics Simulation

The feasibility of the entire approach rests on a recent change in what simulation can do. Classical simulators step one environment at a time on the CPU, on which training one policy per vehicle across a thousand-vehicle bank would be extremely slow. MuJoCo-XLA, the MuJoCo Playground ecosystem [8, 9], Isaac Gym [51] and Brax [52] are GPU-parallel physics engines built on JAX [53] that can step thousands of environments simultaneously on one GPU. This turns population-scale training from a cluster job into a single-GPU job, and the maturity of the JAX ecosystem [54] makes this practical without hand-crafted environments. The scale of this experiment is a direct consequence of that framework.

The sim-to-sim validation uses Stonefish [16], a marine-robotics simulator developed in part by members of the supervisory group and integrated through ROS2. A recent comparative review situates it among ROS-compatible underwater simulators [55]. It is a genuinely independent engine, using a different rigid-body backend, a different hydrodynamics implementation and double-precision CPU integration (64-bit floating point) rather than the single-precision GPU integration (32-bit floating point) used in training. Thus, the two are not numerically equivalent, and Chapter 4 documents a regime, high torque-to-inertia vehicles at the training timestep, in which they diverge sharply.

2.7 Synthesis and Research Gap

Combining all research areas together reveals a gap at their junction which is that Underwater vehicle control is effective but requires per-vehicle model identification and retuning [5, 34]. Teacher–student distillation removes that per-vehicle requirement for aerial vehicles [10], and history-based policies are a known method to get the implicit

system identification such distillation depends on [15]. Fast reinforcement learning for a *single* underwater vehicle is solved [11]. What is missing is the combination: no work distils across a *population* of underwater vehicles, where the dominant off-nominal parameter is buoyancy, an identity parameter with no aerial analogue, and where the distillation literature offers no guidance on how such an identity parameter must be handled. Also, as Chapter 4 shows, handling it as ordinary domain-randomisation noise silently corrupts the training signal.

A second gap exists which this project addresses slightly more. The population-distillation literature establishes that the approach *works*; it does not establish how much of its benefit survives comparison with a single-policy randomisation baseline matched both in evaluation protocol and in the axes randomised. In the underwater domain no such comparison exists at all. This matters greatly because the required computation between both varies immensely. Indeed, the pipeline takes more than 1000 times longer than the randomised policy, so even a modest advantage must be demonstrated rather than assumed. The present work set out to fill the first gap and, in filling it, was obliged to confront the second.

The intersection also exposes why the underwater case is not a re-run of the aerial one. The aerial result exploits thrust-to-weight, a continuous monotonic parameter a policy can estimate from how briskly the vehicle accelerates. Buoyancy is superficially analogous but structurally different. Indeed, the constant force offset must be nulled even at rest, interacts with the pendulum righting moment through the centre-of-buoyancy–centre-of-gravity separation, and renders a vehicle physically uncontrollable once the required hold thrust exceeds the actuator limit.

The dependent variable throughout is tracking error, measured as position and orientation RMS. The independent variables are vehicle mass, hull size and inertia, thrust and torque limits, buoyancy ratio, added payload, and external disturbance force and torque.

2.8 Research Aim, Objectives and Questions

The aim and objectives O1–O5 stated in Chapter 1 are carried forward unchanged. From the two gaps synthesised above, four research questions follow; these frame the logical progression of the remainder of the dissertation:

RQ1 Can a single recurrent policy match per-vehicle specialist controllers across a 5–150 kg AUV bank without retuning, and does it exceed them under off-nominal conditions?

RQ2 How must vehicle-identity parameters such as buoyancy be handled during teacher–student distillation, and what is the measurable cost of handling them incorrectly?

RQ3 Does the resulting adaptation generalise zero-shot to conditions never trained (added payload, external disturbance), and does it survive transfer to an independent physics engine and a physically different vehicle?

RQ4 How does the distilled policy compare against a matched single-policy domain-randomisation baseline trained under the identical environment and evaluated under the identical protocol, and which of the population pipeline’s advantages survive that comparison?

A corresponding directional hypothesis for RQ1 is that a recurrent student will match the specialists on nominal physics and exceed them off-nominal, because the specialists cannot adapt to conditions outside their fixed training configuration whereas the student can. The directional hypothesis for RQ4, held at the outset of the study and derived from the host group’s prior experience, was that the randomised baseline would be substantially inferior across the buoyancy range; Chapter 4 reports that this hypothesis was not supported. Chapter 3 restates these questions and justifies the methodology designed to answer them; Chapter 4 is organised around them; and Chapters 5 and 6 answer and close them in turn.

3 Methodology

Chapter 2 established two gaps: no prior work distils a population of 6-DOF underwater vehicles into a single adaptive policy, and no prior work in this domain compares such a policy against a matched single-policy randomisation baseline. This chapter describes the methodology used to close them. It justifies each decision from the research questions and is assessed against its fitness to answer the four questions given Chapter 2: cross-vehicle performance (RQ1), identity-parameter handling (RQ2), generalisation and transfer (RQ3), and comparison against a domain-randomisation baseline (RQ4).

3.1 Research Philosophy and Strategy

To answer the research questions, a quantitative simulation-based approach was chosen. This is necessary because testing a controller across hundreds of design variations, spanning a thirty-fold range in vehicle mass, would be practically impossible with physical prototypes. Using controlled simulations also guarantees exact repeatability for single-variable comparisons. The main drawback is the reality gap. While sim-to-sim transfer across an independent engine and a distinct vehicle mitigates this (Section 3.9), the lack of real-hardware validation remains a noted limitation (Chapter 6).

3.2 Simulation Environment

The training environment models a 6-DOF underwater vehicle as a free-joint rigid body in MuJoCo-XLA [8], built on JAX [53] with networks in Flax [56] and optimisation in Optax [57], integrated by fourth-order Runge–Kutta at a 0.02 s timestep for a 50 Hz control rate. On top of the rigid-body dynamics it applies a hydrostatic wrench: weight $W = mg$ at the centre of gravity, buoyancy $B = r mg$ at the centre of buoyancy, with an optional offset between the two.

$$\mathbf{f}_{\text{hs}} = (r - 1) mg \hat{\mathbf{z}}, \quad \boldsymbol{\tau}_{\text{hs}}^b = \mathbf{r}_B \times \mathbf{R}^\top(r mg \hat{\mathbf{z}}) - \mathbf{r}_G \times \mathbf{R}^\top(mg \hat{\mathbf{z}}), \quad (2)$$

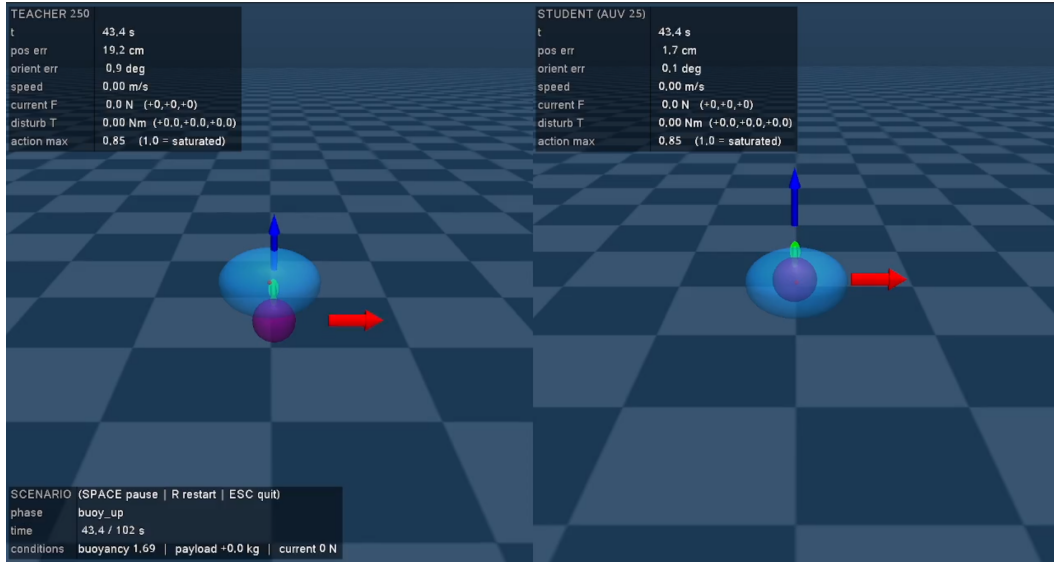
where $r = B/W$ is the buoyancy ratio, $\hat{\mathbf{z}}$ world-up, $\mathbf{R}$ the body-to-world rotation and $\mathbf{r}_B, \mathbf{r}_G$ the body-frame centres of buoyancy and gravity. Both later controllability bounds come from this one expression: the force term drives the buoyancy limit, and the torque term, non-zero only when $\mathbf{r}_B \neq \mathbf{r}_G$, is the pendulum the centre-of-gravity arm introduces. An optional disturbance wrench representing an ocean current is applied alongside it.

Two conventions are fundamental. The observation is twelve-dimensional: body-frame position error (3), axis–angle orientation error (3), body-frame linear and angular velocity (6). The action is six-dimensional, tanh-squashed to $[-1, 1]$, its three force and three torque components each scaled by the vehicle’s own limits. Thus, the policy commands

a *fraction of available authority* rather than an absolute wrench, which is what allows a control law to transfer between vehicles of very different scale. The reward is a weighted negative quadratic in the tracking errors and the command,

$$R = \kappa \left[-w_p \|\mathbf{W}\mathbf{e}_p\|^2 - w_a \|\mathbf{e}_a\|^2 - w_u \|\mathbf{a}\|^2 - w_v \|\mathbf{v}\|^2 - w_m \|\mathbf{a} - \bar{\mathbf{a}}\|^2 - w_r \|\mathbf{a} - \mathbf{a}_{t-1}\|^2 \right], \quad (3)$$

where $\mathbf{e}_p$, $\mathbf{e}_a$ are the position and axis-angle orientation errors, $\bar{\mathbf{a}}$ a moving average of recent commands and κ a return scale. The last three terms are action regularisation: without them a policy reaches low tracking error while chattering the thrusters.



(a) MJX — training and evaluation



(b) Stonefish — sim-to-sim target

Figure 1: The two simulation environments. (a) The MJX dual viewer, running a specialist teacher (left) and the distilled student (right) on the *same* body at the same instant under a buoyancy ratio of 1.69: the teacher holds 19.2 cm, the student 1.7 cm. (b) A BlueROV2-Heavy in Stonefish, the four vertical lines being the dive thrusters' force vectors.

Training and the bulk of evaluation take place in the MJX environment, where a vehicle is an ellipsoid with the sampled mass, dimensions and actuation limits. Due to time limitations we were unable to evaluate the method directly into the robot. We utilise as proxy a sim2sim evaluation strategy using Stonefish [16], an independent marine-robotics engine, driving a detailed BlueROV2-Heavy through its eight physical thrusters (Figure 1). It is important to keep in mind that the visual representation is simply for a nicer display. During training nothing about the target vehicle’s geometry, appearance or thruster layout are present.

3.3 AUV Bank

The population is a sampled hardware bank of one hundred distinct bodies. Mass is drawn log-uniformly over 5–150 kg, so each decade receives equal coverage rather than the sample being dominated by the heavy end. From the sampled mass, hull semi-axes, inertia tensor and thrust and torque limits are derived by mass-scaling laws (length as $\text{mass}^{1/3}$, inertia as $\text{mass}^{5/3}$, thrust as $\text{mass}^{2/3}$, torque as mass), with per-axis jitter so vehicles of similar mass still differ in shape and actuator authority.

3.4 Teacher Policy Training

One PPO teacher is trained per hardware configuration. PPO constrains each update with a clipped surrogate objective,

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}, \quad (4)$$

whose min makes the objective pessimistic, so a single large advantage cannot drive the policy far from the one that collected the data. Because actions must lie in $[-1, 1]$ the Gaussian sample is squashed, $a = \tanh(u)$, requiring the change-of-variables correction

$$\log \pi(a) = \log \mathcal{N}(u; \mu_\theta, \sigma_\theta) - \sum_i \log(1 - \tanh^2(u_i)); \quad (5)$$

omitting it silently optimises a different objective. Each teacher controls one fixed morphology for its entire training lifetime. The principal hyperparameter is the number of parallel environments, 4,096, with 512-step episodes and 301 episodes per teacher (except when early-stopping is used and validated). This batch size justified empirically in Section 4.3.

Checkpoint selection matters because a converged policy can still drift late in training: each teacher’s checkpoints are ranked by the sum of the ranks of their whole-episode position and orientation RMS and the lowest selected. Whole-episode RMS is preferred

over final-step error, which a policy overshooting throughout but ending near the target can luck into; rank-sum over a weighted sum, because the two errors are not on comparable scales.

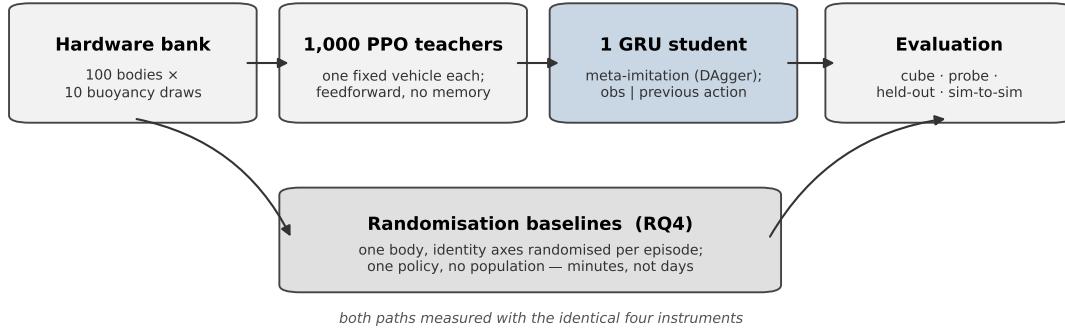


Figure 2: The pipeline and its control condition. The upper path is the population method: one specialist per vehicle, distilled into a single recurrent policy. The lower path is the baseline of Section 3.8, which uses the same environment and evaluation, but with one policy trained on one body with the identity parameters randomised per episode instead of a population of specialists.

3.5 Student Policy Distillation

The teachers are distilled into a single recurrent student built around a gated recurrent unit. Its input at each step is the concatenation of the current observation and previous action. Its output is tanh-squashed to match the teachers’ action bound. Distillation follows a meta-imitation, DAgger-style scheme [14]: the teacher supplies deterministic-mean labels and the student reproduces them under a backpropagation-through-time mean-squared-error loss. After a ten-epoch teacher-driven warm-up the student acts on its own while still labelled by the teacher, training on the states it will actually visit and addressing the distribution shift DAgger identifies.

Recurrence is chosen by argument rather than convention. The parameters distinguishing one vehicle from another are not observable in the twelve-dimensional observation, and the policy is never told which vehicle it controls. Thus, a memoryless policy cannot behave differently for a light neutrally buoyant vehicle and a heavy positively buoyant one presenting the same instantaneous error and velocity, whereas a recurrent policy can. Networks are deliberately small: hidden dimension sixteen, with a sixty-four variant tested to rule out a capacity bottleneck.

3.6 Domain Randomisation Design

This is the most consequential design decision in the study and where RQ2 is operationalised. Two categories of parameter are distinguished and handled oppositely.

Per-episode *nuisance* parameters such as mass, inertia and gravity jitter, are re-sampled every episode, correctly, because the teachers were themselves trained under that jitter. The *identity* parameter, buoyancy ratio, is different in kind: drawn once per teacher and fixed for its lifetime, exactly as physical mass would be. It is sampled at 70% of *that vehicle's own* controllability span rather than a flat range shared across the bank, because a heavy low-margin hull and a light high-margin one tolerate very different absolute swings. That span follows from Equation 2: the net vertical imbalance is $(r - 1)mg$, nullable only while the vertical thrusters can oppose it,

$$|(r - 1) m g| \leq T \quad \Longleftrightarrow \quad |r - 1| \leq \frac{T}{mg}, \quad (6)$$

so the controllable half-span about neutral is $T/(mg)$. A flat absolute range would therefore be too easy for some vehicles but physically impossible for others.

Because one body can host several independent buoyancy draws, the hundred-body bank is expanded to one thousand teachers by drawing ten variants per body. Since each body's variants are independently sampled and the student has no privileged observation of body identity, no discrete grid exists for it to memorise. Hydrodynamic drag is *not* randomised which has a direct consequence for how distillation must be run. During distillation each teacher must be simulated at the fixed buoyancy it was actually trained under, since re-randomising it every epoch trains the student to imitate a specialist evaluated on physics that specialist never saw. Chapter 4 shows this condition was initially violated and quantifies the cost.

Two further identity axes were added later, giving the four training *arms* of Table 2. The **centre-of-gravity offset** reproduces the pendulum restoring moment of a bottom-heavy hull and obeys the same authority-bounding principle: an offset d demands a righting torque of order mgd at large tilt, feasible only while

$$m g d \leq \tau_{\max} \quad \Longleftrightarrow \quad d \leq \frac{\tau_{\max}}{mg}. \quad (7)$$

Because semi-axes scale as $m^{1/3}$ while $\tau_{\max}$ scales as m , a flat geometric fraction for d fails at the heavy end. Thus, d is instead capped relative to each vehicle's own torque authority. The **actuation lag** models thruster spin-up and pipeline delay as a first-order filter on the applied wrench,

$$\tau_t = \alpha \mathbf{a}_t + (1 - \alpha) \tau_{t-1}, \quad \alpha = \frac{\Delta t}{\tau_{\text{lag}} + \Delta t} \approx 0.29 \quad \text{for } \tau_{\text{lag}} = 50 \text{ ms}, \Delta t = 20 \text{ ms}. \quad (8)$$

This makes two states with identical error and velocity evolve differently depending on τ_{t-1} . Thus, the environment would not be Markovian anymore. Therefore, the

lagged wrench is appended to the observation, restoring the Markov property and giving eighteen dimensions for those arms.

Table 2: The four training arms. Every arm varies buoyancy and body geometry, differing in which further identity axes are randomised; each has a matched baseline in Section 4.12.

Arm	Adds	CoG offset	Start attitude	Actuation lag	Obs. dim
A	— (buoyancy only)	—	identity	—	12
B	attitude realism	yes	$\pm\pi/2$	—	12
C	both	yes	$\pm\pi/2$	$\tau = 50$ ms	18
D	actuator realism	—	identity	$\tau = 50$ ms	18

3.7 The Domain-Randomisation Baseline (RQ4)

The control condition for RQ4 is a single feedforward PPO policy trained on *one* vehicle (the BlueROV2-scale body at the centre of the bank) with the buoyancy ratio re-randomised every episode across that vehicle’s controllable span. It is matched in implementation with the same environment, physics, timestep, observation, action normalisation, reward, network sizes, optimiser and batch size but differing in that buoyancy is a per-episode draw and there is one policy rather than a population. On its own it is *not* matched in randomisation breadth, varying buoyancy on one body where the student arms were distilled over populations additionally varying geometry, centre-of-gravity offset, start attitude and actuation lag. Rather than leave that as a limitation, **three further baselines** were trained, one per remaining arm, each copying that arm’s randomisation set verbatim: offset and $\pm\pi/2$ start (Arm B); actuation lag, and hence the same eighteen-dimensional observation (Arm D); and all four axes together (Arm C). All four are evaluated in every instrument the students appear in except the thruster-fault sweep.

The early-stopping criterion watches whole-episode RMS while the reported metric is settled hover. Thus, the Arm B baseline was retrained without early stopping over the full budget to confirm its raised error floor is a capability limit rather than an artefact of the stopping rule. Cost is reported in teacher-trainings rather than wall-clock hours, for comparability across hardware. The two configurations ran on different devices; the resulting bias works against the baseline.

This baseline was implemented late, initially to generate an illustrative figure rather than as a planned control. The project had assumed from the start that this baseline would fall short, rather than putting that assumption to the test.

3.8 Evaluation Protocol

Four instruments are used, each targeting a different question (Table 3). Three design choices apply across all of them. Reported values are *settled* error i.e. the mean over each episode’s final 40%, because whole-episode RMS is transient-dominated and cannot resolve steady-state hold. Where a *success rate* is reported, a run counts as successful when both settled errors fall below 0.05 in their own units, 0.05 m and 0.05 rad ($\approx 2.9^\circ$). All stressor magnitudes are scale-invariant, expressed as fractions of each vehicle’s own controllability span, mass or actuator limits, so the identical schedule is a fair test on a 5 kg and a 150 kg vehicle alike. Finally, every policy is driven through the identical timeline on the same body, under fixed seeds and deterministic replay.

Table 3: The four evaluation instruments.

Instrument	Question	Design
Condition cube (primary)	Does the policy beat the specialists, and where?	7 vehicles $\times$ 14 isolated conditions $\times$ every policy, from hover and tracking through buoyancy, payload, CoG offset and disturbance to all-at-once. A 17-phase, 102 s scripted scenario gives the time-resolved view.
Hidden-state probe	<i>How</i> does it adapt? Inference or reaction?	Linear ridge probe from settled hidden state to a physical parameter, fit on 700 teachers and scored on 300 held out (after [10]), with label-shuffle and random-hidden controls.
Held-out banks	Does <i>control</i> , not just representation, generalise?	Four fresh 100-body banks at an independent seed: in-range, out-of-range-high (150–300 kg), out-of-range-low (2–5 kg) and anisotropic (aspect ratios to 5:1).
Sim-to-sim	Does any of it survive a different engine and vehicle?	BlueROV2-Heavy in Stonefish [16] via a ROS2 bridge, with ArduSub thrust allocation and a body-frame conversion. A seven-configuration thruster-fault sweep adds an out-of-distribution actuator test.

3.9 Ethical, Data-Management and Sustainability Considerations

The study is purely computational, involving no human participants, animal subjects or primary data collected from people, and so raises none of the consent, confidentiality or vulnerability concerns the university’s ethics process addresses. All data are simulation outputs (checkpoints, telemetry and metrics) held in a version-controlled repository alongside the scripts that produced them, so every reported number is traceable to its run.

Sustainability is material rather than a formality here, because the central finding of

Chapter 4 is partly an efficiency finding: the population design trains one thousand policies and distils them where the baseline of Section 3.8 trains one, a difference of roughly three orders of magnitude in training compute that Chapter 5 argues must enter the engineering judgement, not merely the accuracy comparison.

4 Findings and Data Analysis

This chapter presents the measured results in the order the work was carried out, because that order is itself part of the finding: the sample and validity measures, the teacher population, the distillation failure and its diagnosis (RQ2), the teacher–student comparison and mechanism probe (RQ1), generalisation to unseen vehicles and engines (RQ3), and finally the late baseline comparison that revised the study’s own premise (RQ4). Every figure is generated directly from the run data.

4.1 Description of the Sample

The sample is the hardware bank of one thousand teacher configurations: one hundred physical bodies, each with ten independent buoyancy draws (Figure 3). Mass spans 5.1–143.3 kg log-uniformly (median 24.8 kg), roughly 28 $\times$, with the reference BlueROV2-Heavy at 11.75 kg below the median. Thrust limits span 48.0–575.3 N and per-axis torque limits 3.7–153.4 N m, both rising with mass as designed. The buoyancy ratio spans 0.12–1.84 (median 1.01), its spread deliberately wider for low-mass high-margin vehicles. This diversity is what makes the bank a test of generalisation rather than interpolation over near-identical vehicles.

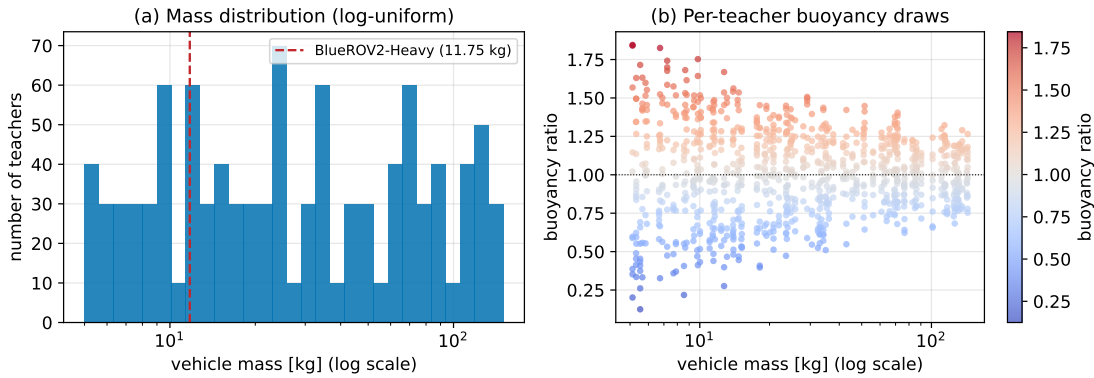


Figure 3: The sampled hardware bank of 1,000 teacher configurations. (a) Log-uniform mass over 5–150 kg. (b) Buoyancy drawn relative to each vehicle’s own controllability span, giving wider spreads for light high-margin vehicles.

4.2 Validity and Reliability

Several measures secure internal validity. All training and evaluation use fixed seeds and deterministic replay, and in every policy-versus-policy comparison the policies are driven through the identical timeline on the same physical body, so differences are attributable to the controller rather than the test conditions. Checkpoint selection follows the rank-sum rule of Chapter 3. A three-seed sweep of the distillation gives final position errors of 1.65, 1.55 and 1.92 cm which is a spread ten times smaller than the effects

reported below, so the headline result isn't just a lucky seed. Two limitations are stated rather than left to be found: the sim-to-sim transfer is a single transfer to one vehicle on one engine, and the baseline comparison of Section 4.12 replicates only two of its four baselines over seeds. A third limitation was that the first baseline had been randomised over only one identity factor, whereas the students were trained across several. This was addressed by training three further axis-matched baselines, rather than leaving the mismatch unresolved.

4.3 The Teacher Population

The teachers converged across the whole bank. Figure 4 shows each selected teacher's final-step tracking error, taken from the training-time evaluation rollout: a median position error of 2.30 cm (IQR 1.6–3.4 cm) and orientation error of 0.62° (IQR 0.44 – 0.87°) across all 1,000 teachers. This is a different quantity from the settled-window error of Section 3.9 and not comparable to the 0.05 success criterion; it is reported because it is what the training logs record for every teacher, and it confirms the specialists are uniformly competent. A visible tail extends well beyond the bulk of the distribution. 5.8% of teachers exceed 5 cm in position error, while no teacher's orientation error rises much above the upper end of its interquartile range. This tail is not a training failure: those teachers were drawn roughly twice as far from neutral buoyancy as the rest (mean $|r - 1|$ of 0.39 against 0.21), placing them near the controllability limit of Equation 6.

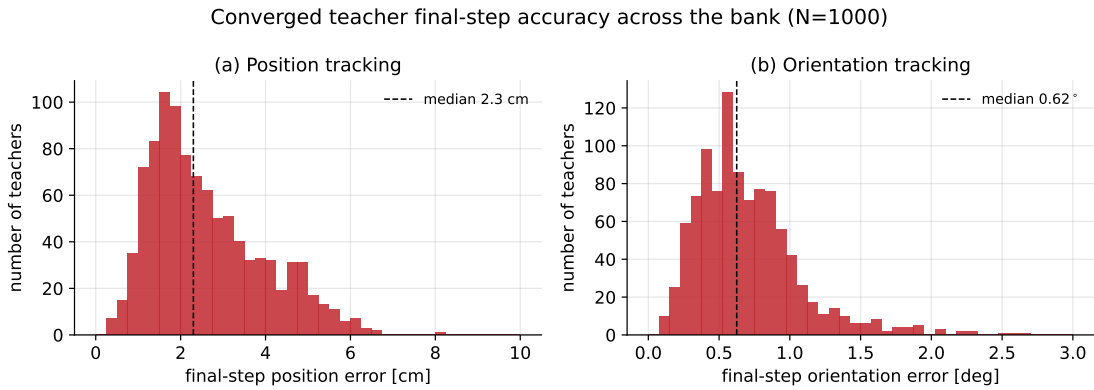


Figure 4: Converged teacher final-step accuracy across the bank ($N = 1000$), from the training-time evaluation rollout. Not the settled-window metric of Section 3.9: median 2.30 cm and 0.62° .

Two secondary observations matter later. Converged error is essentially *flat in mass*: teacher difficulty tracks $|\text{buoyancy} - 1|$ rather than weight, so a heavy vehicle is not intrinsically harder to control once actuation is scaled to it. And training was sensitive to batch size exactly as Section 2.3 predicts: development runs at 64 parallel environments collapsed into local minima while runs at 2,048 or more converged reliably. Since

Tunçay S. (from [11]) ran his trainings with 4096 environments and that the personal laptop could just handle this, that value was used for the whole project.

4.4 The Distillation Plateau

The first attempt to distil a buoyancy-diverse student plateaued but far from the non-buoyancy reference. Imitation loss and position error fell for roughly twenty epochs, then plateaued at approximately 0.045 and 4.18 cm until the 200th epoch. Over the same pipeline, a non-buoyancy reference student reached 0.015 and 0.34 cm which is thirteen times better in position control. Because the plateau held all the way to epoch 200, slow convergence could be ruled out as the cause. The training really hit a plateau and finding the cause became the central investigation of the study.

4.5 Interventions That Did Not Resolve the Plateau

Three interventions, each giving the student more capacity or better information, were tried and failed (Table 4). Increasing capacity sixteen-fold left the plateau essentially unchanged. Appending an accumulated integral-error term did not help neither; nor did a body-frame gravity-direction feature. The last one even made the orientation error 1.5–2 times worse which is consistent with diluting a small network’s capacity rather than adding signal. The decisive observation is that all three structurally different interventions converged to the *same* plateau (Figure 5), showing that the problem is not in the student at all.

Table 4: Interventions on the buoyancy plateau (final-epoch values). The non-buoyancy reference student, distilled under the identical pipeline, is the standard the others fall short of. All three interventions converge to the same ~4.2 cm plateau; only the label fix departs from it.

Intervention	Loss	Pos. err. [cm]	Verdict
<i>Non-buoyancy reference</i>	<i>0.015</i>	<i>0.34</i>	<i>reference</i>
Baseline (hidden 16, re-randomised buoyancy)	0.045	4.18	plateau
Capacity (hidden/latent 64)	0.046	4.14	no change
+ integral-error observation	0.050	4.43	no change
+ gravity-direction observation	0.052	4.46	no change
Buoyancy label fix (hidden 16)	0.020	1.91	resolved

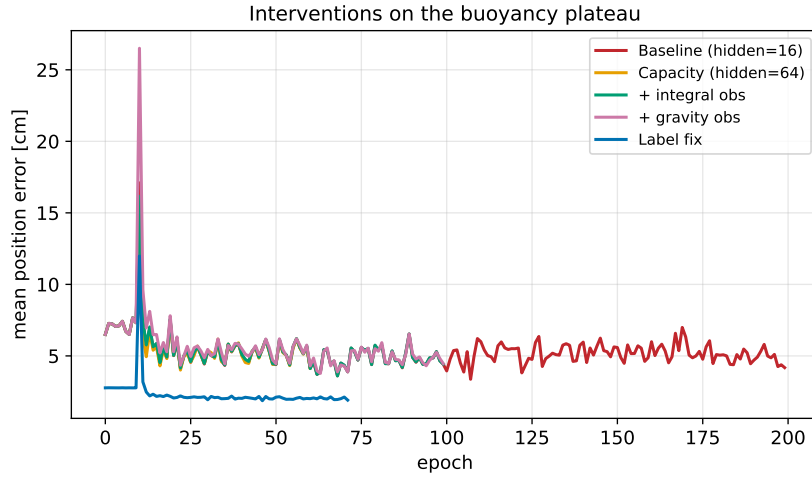


Figure 5: The three interventions against the baseline, over training. That three structurally different changes share a floor is the evidence the limit lay outside the student; only the label fix departs from it.

4.6 The Identity-Parameter Fix and the Controllability Bound

The plateau was caused by a label-consistency error. During student rollout the buoyancy ratio was re-sampled every epoch, uncorrelated with the fixed value the imitated teacher had trained under. Replaying the exact random derivation confirmed the mechanism: teacher 80, trained at buoyancy 1.335, was simulated at 0.166, 0.331, 0.294 and 0.172 across four successive epochs. The student was imitating a buoyancy-1.335 specialist evaluated on buoyancy-0.166 physics. The label was not only noisy but wrong, which is why no student-side capacity or observation could repair it.

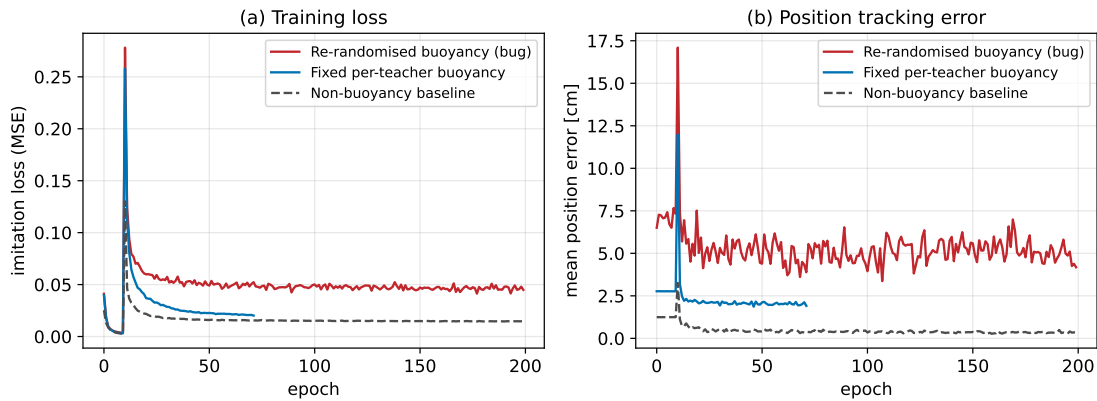


Figure 6: The buoyancy label-consistency fix: (a) imitation loss and (b) position error for the re-randomised student (red), the fixed-per-teacher student (blue) and the non-buoyancy reference (grey).

Holding each teacher’s buoyancy fixed at its trained value (verified: 1.335 every epoch after the fix, matching its manifest) removed the plateau. The corrected student reached

approximately 0.020 loss and 1.91 cm position error, roughly halving both (Figure 6). This single change which did not affect network, observation nor hyperparameter, but only label consistency, is the largest single improvement in the study and the answer to RQ2. It also explains why earlier non-buoyancy students escaped the same latent flaw: buoyancy was then randomised within $\pm 1\%$, so any redraw landed back near 1.0. Widening the range turned a dormant bug into a severe one.

The controllability bound of Equation 6 was confirmed both across the bank and behaviourally. The median half-span is 0.65, from 0.29 for the least-margin vehicles to 1.35 for the most (Figure 7). The figure shows the spread that justifies scaling both randomisation and evaluation to each vehicle’s own margin, rather than applying one fixed range to every vehicle. Sweeps of a low-margin vehicle held tracking to a few centimetres through the safe range, degraded to about 11 cm near the hard limit, worsening to roughly one metre once at the limit, a progression recorded in Video V3 (Appendix 4).

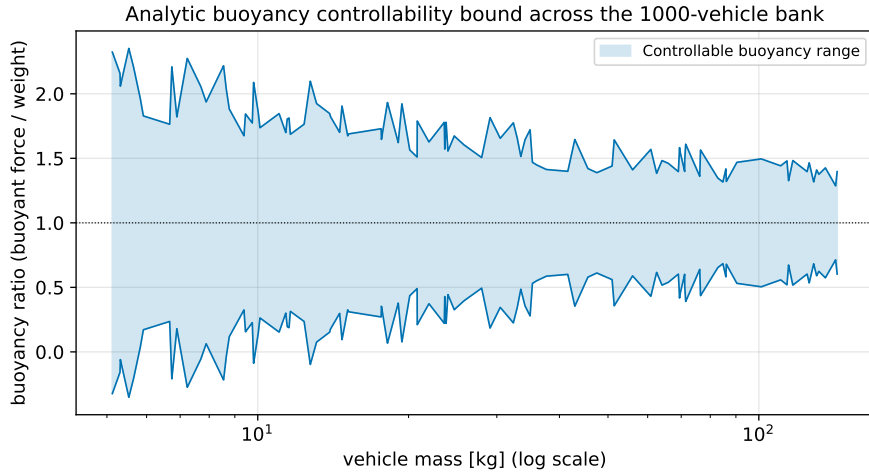


Figure 7: Analytic buoyancy controllability bound: the range each vehicle can hold, half-span $T/(mg)$ (median 0.65, range 0.29–1.35).

4.7 Specialists versus the Distilled Generalist

With consistent labels the central comparison for RQ1 could be made. Figure 8 presents the condition cube for which seven per-class specialists and five distilled students were deployed on seven vehicles spanning the mass range, and tested on fourteen isolated conditions. It shows the settled error averaged over the vehicles for each policy on each task.

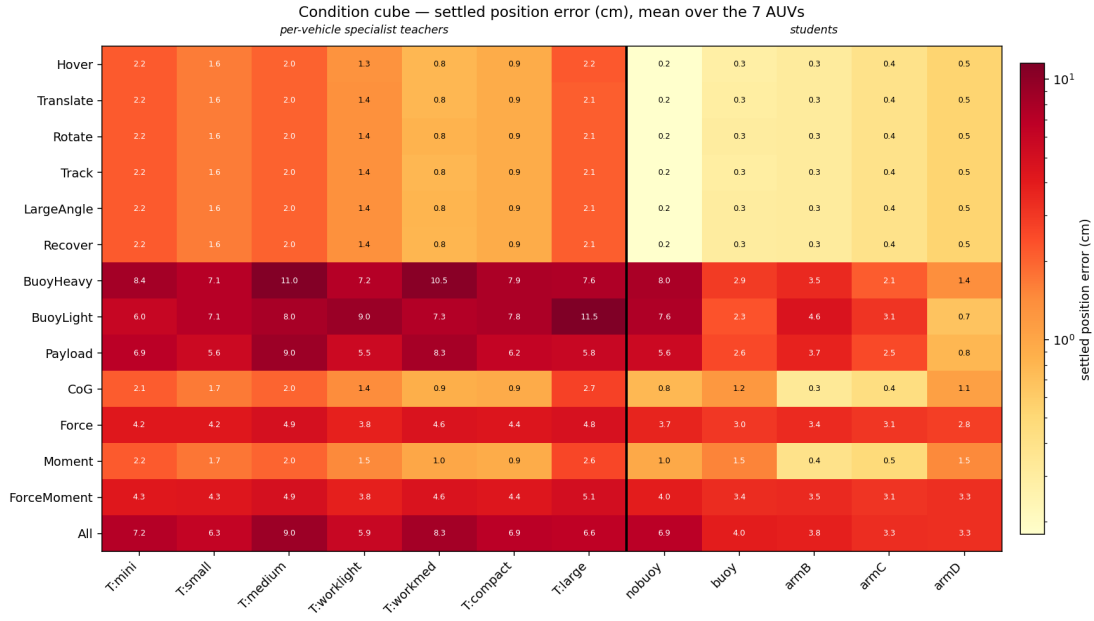


Figure 8: Condition cube: settled position error (cm), mean over seven vehicles, for seven specialist teachers (left) and five distilled students (right).

The pattern is a colour split at the teacher–student divider. As expected, on off-nominal conditions the student greatly outperforms the specialists (7.6–11.0 cm against 1.4–3.5 cm under strong buoyancy, 5.8–9.0 against 0.8–3.7 under heavy payload). Surprisingly, on near-nominal conditions the students are also tighter than the specialists (2.0–2.2 cm and the students at 0.2–0.5 cm).

One column is especially informative, and anticipates Section 4.12. The nobuoy student that was distilled from teachers that saw only $\pm 1\%$ buoyancy variation fails the strong-buoyancy condition at 8.0 cm, indistinguishable from the fixed specialists, despite being architecturally identical to the students that succeed at 2.9 cm. The effect therefore lies in *buoyancy diversity in training*, not in recurrence or capacity alone. Both students share the same architecture, so what differs is what each was exposed to during distillation, not what each is capable of representing.

The time-resolved scenario (Figure 9) tells the same story with a caveat. The complete schedule, with both policies stepping it in lockstep, is recorded as Video E1 (Appendix 4), alongside the other full-length runs. Per-phase position RMS for a specialist against the student was 19.24 versus 3.55 cm under strong upward buoyancy, 11.43 versus 3.30 downward, 8.48 versus 3.06 under heavy payload and 9.75 versus 4.88 under a rotating current, while steady current was comparable. Contrary evidence is reported alongside, and it is not confined to a narrowing of the gap: the student is outright *worse* on four phases, three of which command a *moving* target. That concentration is the informative part. Of the five phases with a moving target, the student loses three, against one of the

nine with a fixed target. The generalist tracks a moving reference less sharply than a specialist tuned to one vehicle, which is the price of a control law serving a thousand. Orientation splits along a different seam: the student is worse on the two large-attitude phases but better on every identity and disturbance phase, by factors of two to five.

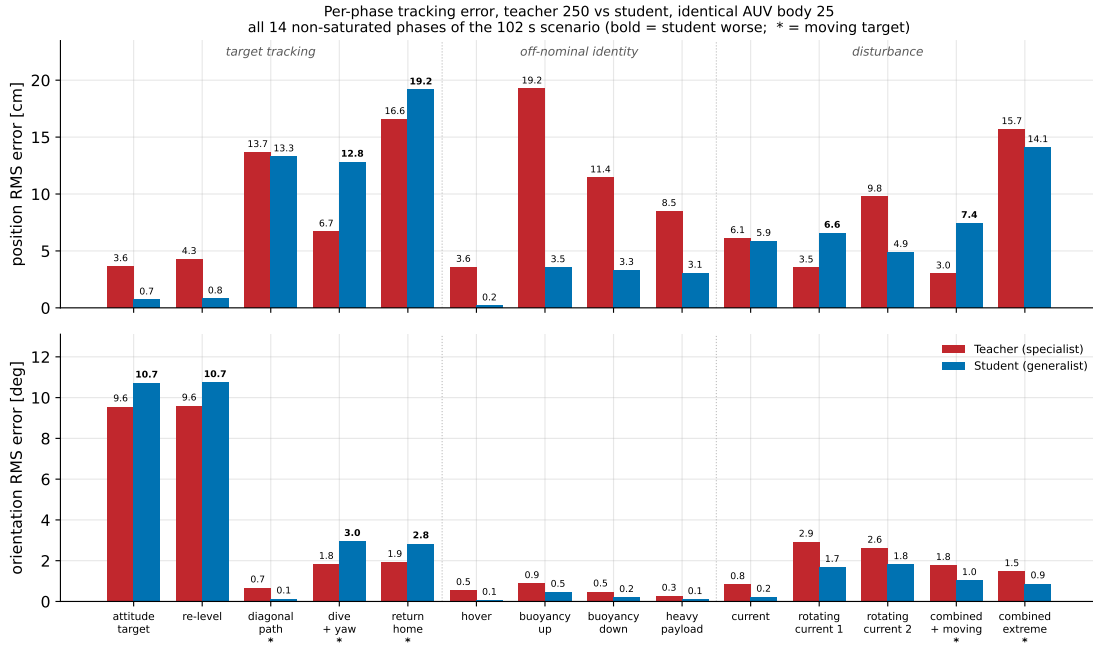


Figure 9: Per-phase position and orientation RMS, specialist versus student, on the identical vehicle and 102 s timeline. All fourteen non-saturated phases are shown, including the four where the student loses (bold). The three large-step phases are excluded: at 17–23% thrust saturation their error measures how long the hull takes to cross the gap, not the control law.

The whole-scenario totals are nearly equal (32.31 versus 31.71 cm), but that number should not be read as a policy comparison. Indeed, three large-step phases occupy 16% of the timeline yet contribute 90% of the total squared error. In those phases, both policies are already commanding maximum thrust, so the error mostly reflects how long the vehicle physically takes to cover the distance and not which controller is driving it. Excluding those three, the pooled RMS across the remaining twelve phases separates as the per-phase pattern implies: 11.08 cm for the specialist against 9.29 cm for the student. The result is a trade rather than a clean sweep: robustness to the vehicle is bought at the cost of sharpness on a moving reference.

A separate reading of the same cube answers a question about scale transfer. In the teacher $\times$ vehicle cross-transfer grid (Figure 10) the *rows are flat*: a 5 kg mini specialist scores essentially the same on all seven bodies, including the 143 kg one, as on its own, and the boxed diagonal is not special. Body scale, in this normalised-authority formulation, transfers for free. Row-to-row differences actually reflect teacher quality rather than scale mismatch.

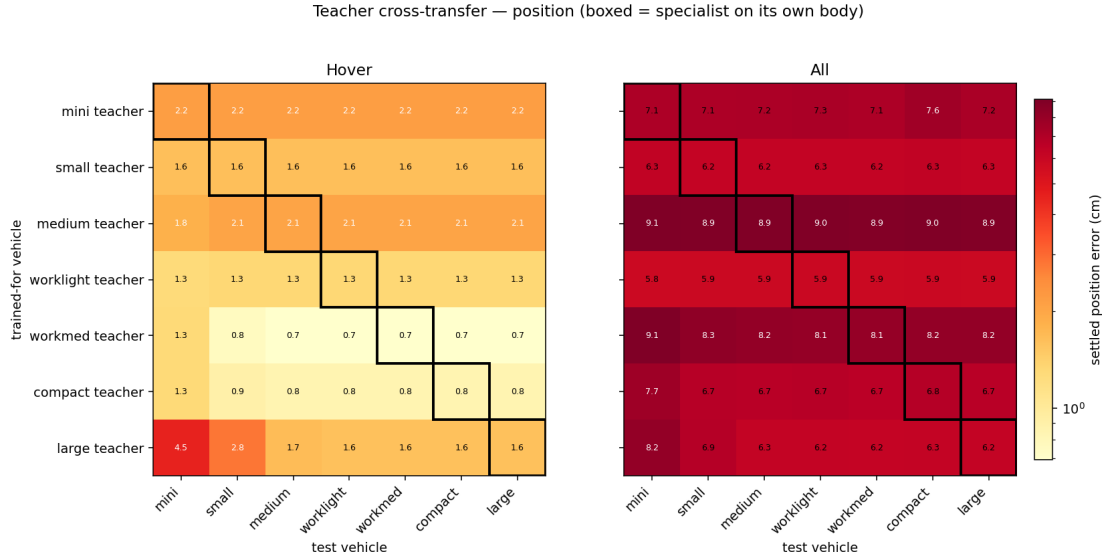


Figure 10: Teacher $\times$ test-vehicle cross-transfer at hover (left) and under combined stress (right); boxed cells mark each specialist on its own body. The flat rows show body scale transfers essentially for free.

4.8 Probing the Mechanism

The evidence that the student *infers* the vehicle rather than memorising the training set comes from probing its hidden state, following RAPTOR’s latent-probe method [10]. For each of the 1,000 teachers the student was rolled out on that teacher’s fixed physics and its settled sixteen-dimensional hidden state recorded; a linear probe (with ridge regularisation) was trained to predict each physical parameter from the hidden state, using 700 teachers for fitting and the remaining 300 for evaluation. Buoyancy is linearly decodable at a held-out R^2 of 0.87 against a training 0.89. This near-equality shows no overfitting despite sixteen features. The derived net vertical imbalance came at 0.65, while mass and thrust reached only 0.02 and 0.04, meaning that bulk body identity is essentially *not* decodable, and indistinguishable from the two controls.

A further target makes the structure of these numbers clear. The quantity the controller actually needs is not the buoyancy ratio but the *fraction of its own thrust* that hovering consumes, $(r - 1)mg/T$. This is same normalised currency the action space uses (Section 3.3). Probing the identical sixteen units for that quantity returns a held-out R^2 of **0.9999**, with training and held-out agreeing to four decimal places and a residual standard deviation of 0.0035 against a range of ± 0.70 . The three targets above are then not independent findings but partial reconstructions of this one: buoyancy reaches 0.87 because r is largely recoverable from the command once the m/T factor is absorbed and the imbalance in newtons reaches only 0.65 because converting back to an absolute force requires the thrust limit, which the network has no reason to represent. Finally, mass and

thrust sit at the control level because they never appear in the policy’s problem except inside that ratio. Additionally, buoyancy alone predicts the net imbalance at $R^2 = 0.41$ across the bank, yet the probe reaches 0.65, so it is demonstrably not routing through buoyancy.

This sharpens the claim of Section 5.2 from *selective* to something more specific: the recurrent state encodes the control-relevant quantity in the units the controller commands. Because the policy commands a fraction of available authority rather than an absolute wrench, the identity parameter it must infer is likewise expressed as a fraction of authority. The normalisation propagates from the action space into the learned representation. That the network settles on this scalar, rather than on the physically named parameters an engineer would choose, is the clearest evidence in this study that the adaptation is genuinely learned rather than imposed by the experimental framing.

Table 5: Linear-probe decodability of vehicle parameters from the GRU hidden state ($N = 1000$, held-out $n = 300$), with the values being the R^2 for each test.

Target	Train	Held-out	Shuffle ctrl.	Rand-hidden ctrl.
Hover command $(r-1)mg/T$	1.00	1.00	0.00	−0.02
Buoyancy ratio	0.89	0.87	−0.03	−0.02
Net vertical imbalance	0.70	0.65	0.02	−0.00
Mass	0.03	0.02	0.01	−0.00
Thrust limit	0.04	0.04	−0.03	−0.02

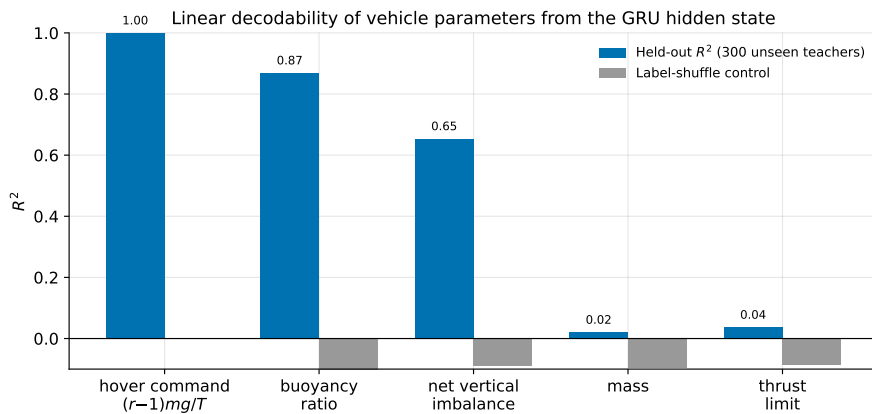


Figure 11: Held-out decodability of vehicle parameters from the hidden state, against the label-shuffle control. The hover command decodes essentially perfectly; buoyancy and the net imbalance are partial reconstructions of it, and mass and thrust sit at the control level.

Applying the fitted probe at every timestep shows the estimate forming *within* the episode (Figure 12). The mean absolute error starts high, at $t = 0$ the hidden state is zero, so every vehicle receives the same guess and decays to near zero over roughly 2.5–3 s.

Converging to their own level in 120 to 150 control steps, with held-out teachers binned by true buoyancy shows a concrete adaptation latency, the dynamic counterpart to the static R^2 . Video V2 (Appendix 4) shows the same re-convergence when the vehicle is exchanged mid-flight with the recurrent state preserved.

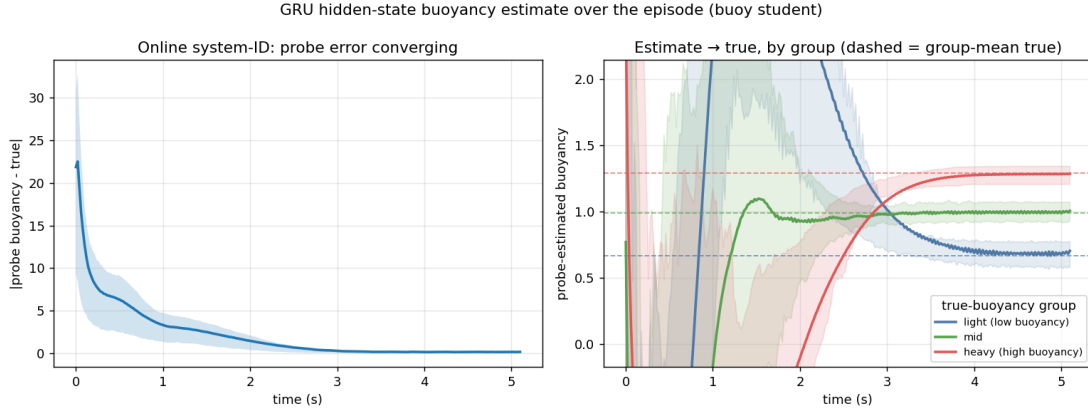


Figure 12: Online formation of the buoyancy estimate. Left: mean absolute probe error over 300 held-out vehicles, decaying over ≈ 3 s. Right: vehicles binned by true buoyancy, each converging to its own level.

Although this evidence shows the information is present and decodable, it does not prove that the control law uses it optimally. This is a limitation the next section addresses.

4.9 Generalisation to Unseen Vehicles

The probe establishes that the *representation* generalises. Whether *control* generalises was tested by evaluating every trained policy, unchanged, on four freshly generated banks of one hundred vehicles each that no policy had ever trained on (Figure 13, Table 6).

Table 6: Held-out generalisation: settled position error (cm) and success rate on four unseen banks. “Train” is the original bank; success is 100% unless shown. Baselines and students are like-for-like here, neither having trained on these vehicles.

Policy	Train	In-range	OOR-high	OOR-low	Anisotropic
DR buoy	0.91	0.91	0.91	0.92 (100%)	0.91
DR cog	1.61	1.61	1.61	1.51 (98%)	1.61
DR armC	1.54	1.54	1.54	1.54 (100%)	1.54
DR delay	1.09	1.09	1.09	1.09 (100%)	1.09
nobuoy	1.35	1.35	1.35	2.58 (74%)	1.36
buoy (Arm A)	1.19	1.19	1.14	2.50 (71%)	1.16
Arm B	2.08	2.10	2.09	2.03 (53%)	2.10
Arm C	1.63	1.62	1.55	1.40 (67%)	1.62
Arm D	0.74	0.74	0.74	0.94 (100%)	0.74

Position and attitude error are essentially *flat* across the training bank, unseen in-range vehicles, vehicles at twice the maximum training mass, and hulls with aspect ratios far outside the training distribution. The policies control brand-new vehicles at training-level precision except the light boundary, where success falls to 53–74% for most arms, Arm D being the standout at 100% everywhere. However, that limit is specific to the distilled policies since *every* randomisation baseline in Section 4.12 holds 100% on all four banks. It is a property of the adaptive architecture rather than of the task. Section 5.4 takes up why.

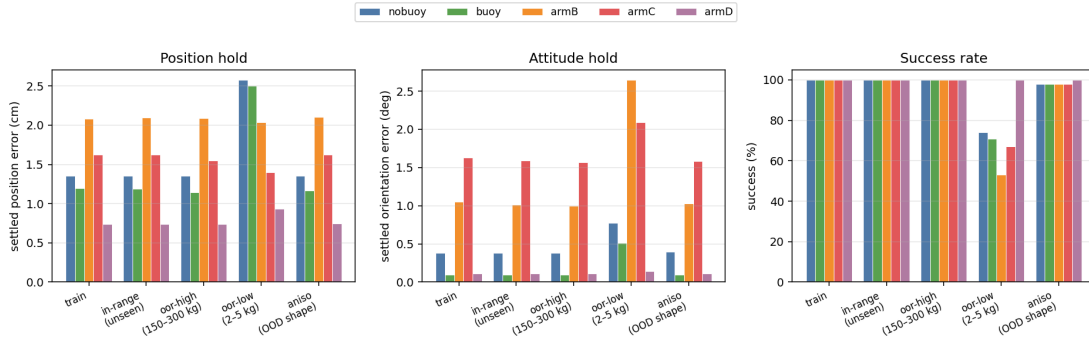


Figure 13: Held-out generalisation across four unseen vehicle banks; flat except at the light-mass boundary.

Position accuracy and success rate do not track together at the light boundary. Arm B and Arm C hold competitive position accuracy there (2.03 cm and 1.40 cm) while their success rate falls to 53% and 67%. Figure 13’s attitude panel shows why, with orientation error spiking sharply for these arms at OOR-low. Success requires both position and orientation to stay within threshold, so a policy can fail the task while still tracking position well, simply by losing attitude control under low buoyancy.

The same test on the fixed specialists is worth recording: a 5 kg mini specialist holds 2.2 cm and 100% success on the 150–300 kg bodies *and* the anisotropic shapes, identical to its own hull. Indeed, near the end, Tunçay S. from the team seemed to have noticed that his Fast6DOF pipeline was already quite geometry robust. Which at first was considered a hard axis became actually almost free thanks to his method.

4.10 Extending the Approach to Further Identity Axes

Sim-to-sim testing (Section 4.11) exposed attitude dynamics absent from training, motivating the further arms of Table 2. The centre-of-gravity arm’s first attempt exposed a design flaw: sampling the offset as a flat fraction of geometry ignores torque authority, so the demanded righting torque, growing with mass, exceeded the roll and pitch limit for heavier classes. The ratio of demanded to available torque rising from

0.75 for the lightest class to 2.11 for the heaviest, made it physically impossible for some vehicles to right themselves. The heaviest class settled at 40 cm and 47° with zero success: uncontrollable by construction. Capping the offset to each vehicle's own torque authority resolved it, every class in a retrained bank settling to roughly 3 cm and $1.1\text{--}1.4^\circ$ with the expert rate rising from 33% to 90.3%. The actuation-lag arm made the environment non-Markovian for a memoryless teacher, so the lagged wrench had to be appended to the observation before teachers would converge at all. This fix consisted of a state-augmentation argument from control theory, not a learning-side adjustment. With it, a per-class pilot converged across the whole mass range (Figure 14) where the unaugmented formulation stalled near 50 cm.

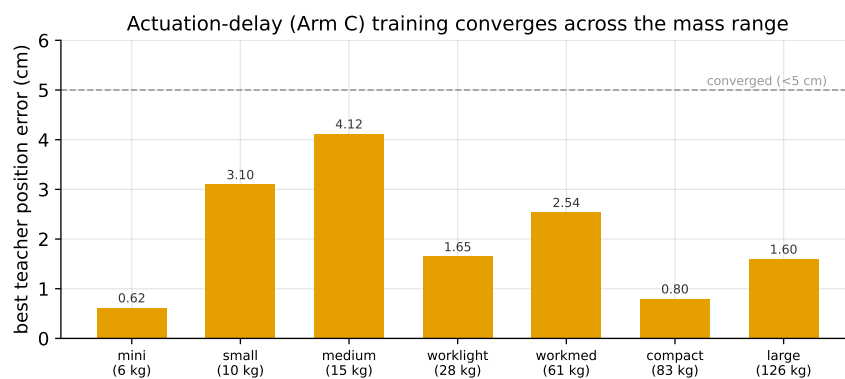


Figure 14: Arm-C per-class pilot after the Markov observation augmentation: best teacher position error by mass class.

4.11 Sim-to-Sim Transfer and Actuator Faults

The trained student was transferred to a BlueROV2-Heavy in Stonefish, an independent engine and a physically different vehicle (Figure 1b). Two bridging steps had to be correct first: a thrust allocation converting the 6-DOF wrench to eight fixed thrusters, for which the validated ArduSub vectored mixer was used rather than a geometry-derived matrix that initially drove the vehicle the wrong way; and a frame conversion between the engines' opposite body conventions, whose omission was masked at low thrust but caused divergence at high thrust.

With those correct, position transferred, but how well depended sharply on which arm the student came from, and the pattern is itself the result. Driving all five students through the identical trajectory, settled hover error was 6.20 cm for the buoyancy student, 4.48 for the centre-of-gravity student and 3.08 for the combined arm, but only **1.03 cm** for the actuation-delay student. The vertical channel telling the same story more starkly, the delay-free students bobbing over an 8.9–12.0 cm depth range against 2.4 cm for the delay-trained one. That ordering is not incidental. Stonefish has a real actuation latency of roughly 50 ms which MJX does not model at all. This is a policy distilled without it

carries a vertical loop gain that is stable in training and marginally stable on the target engine. The resulting limit cycle is precisely the failure the delay arm was introduced to remove, and is visible in Video V5 (Appendix 4). Another problem was that attitude transferred less cleanly as roll initially oscillated at 18.5° standard deviation. This was a ROS2 message-timing artefact and got reduced to 0.4° by tightening the setpoint queue and a graphics-safe concurrent-execution fix. This was a pipeline fix, not a control improvement but a real physical attitude limitation remained and is analysed in Chapter 5.

The attitude comparison across both engines is shown in Figure 15: the centre-of-gravity arm holds tilt more tightly than the buoyancy arm in both, but the Stonefish panel also shows pitch degrading far faster than roll for both policies.

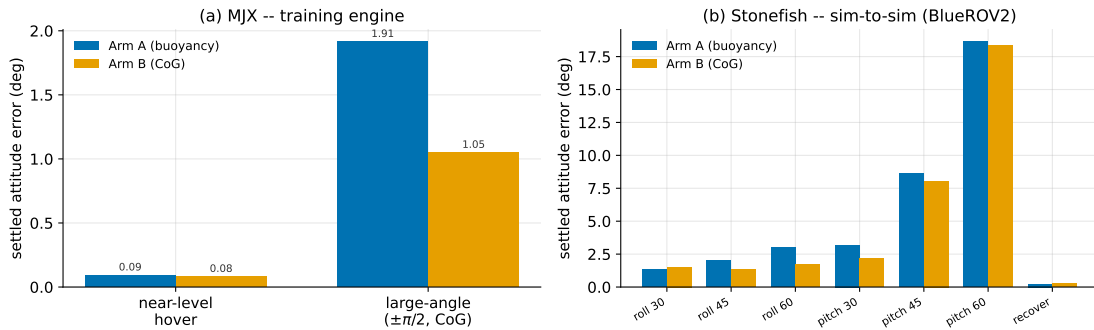


Figure 15: Settled attitude error, buoyancy versus centre-of-gravity student. (a) MJX regimes; (b) Stonefish hold phases on the BlueROV2.

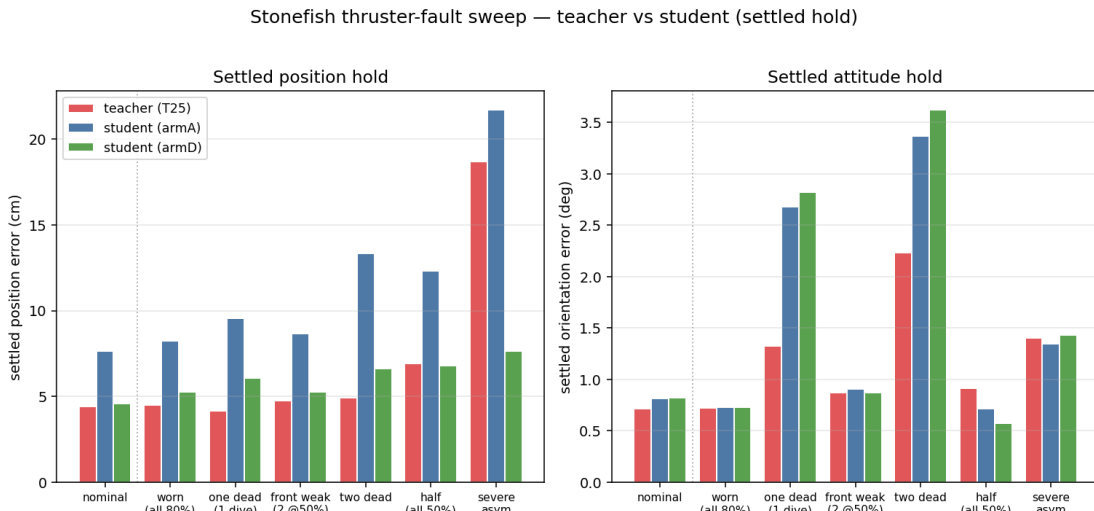


Figure 16: Stonefish thruster-fault sweep: settled position (left) and attitude (right) for a specialist teacher and two students across seven actuator configurations, ordered by severity.

The thruster-fault sweep provided an out-of-distribution actuator test no policy trained for (Figure 16). Across seven configurations from mild wear to severe asymmetry, the

delay-trained Arm D student held settled position markedly better than the delay-free Arm A on every fault, and under severe asymmetry outperformed both Arm A and the specialist (7.6 cm against 21.7 and 18.7 cm), as recorded in Video V6 (Appendix 4). No randomisation baseline was run in this sweep, so it compares students against a specialist only. A methodological note is recorded because it nearly produced the opposite conclusion: Arm D's action trace is saturated and visibly chattering, reading as poor control, yet its settled hold is the tightest of the three. In this case, saturation being the signature of the delay-trained high-gain law, not of instability.

4.12 The Baseline Comparison

Aiming for some nice plots to compare my pipeline to the DR baseline, I trained A single feedforward policy on one vehicle with buoyancy randomised per episode. The baseline had already been tested and branded unsuccessful but did not behave as expected after converging in only 635 seconds.

Figure 17 sweeps buoyancy on one vehicle and separates three response shapes cleanly. The fixed specialist traces a sharp **V**: tight only near its own trained buoyancy and degrading to 11.4 and 19.3 cm at the extremes showing the failure that motivated the entire study. The distilled student traces a shallow **bowl**, 0.26 cm at neutral rising to 1.5–3.3 cm. Surprisingly, the randomised baseline traces a *very similar bowl*: 0.58 cm at neutral, 3.0–3.3 cm at the extremes. Both adaptive approaches decisively beat the specialist while not decisively differing from each other. Video V7 (Appendix 4) drives the baseline and a student under identical commands.

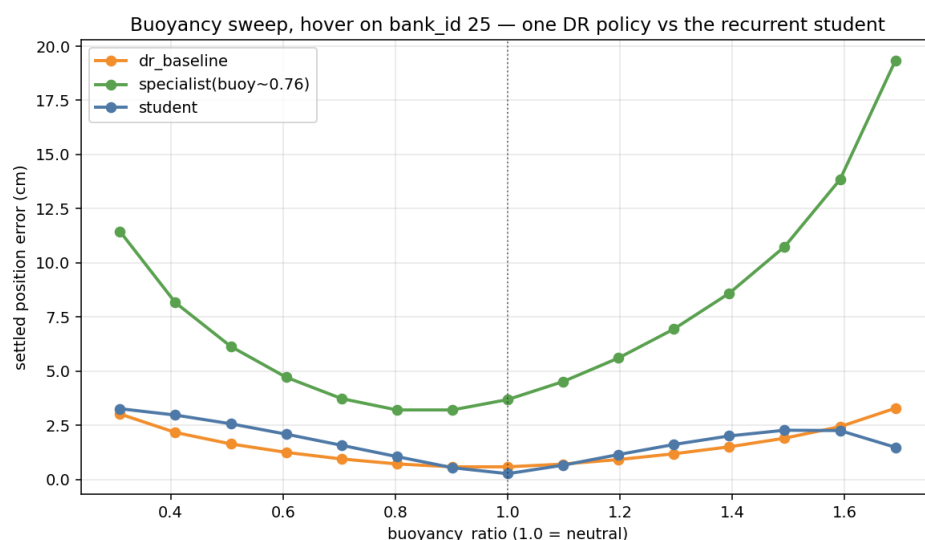


Figure 17: Settled position error against buoyancy on one vehicle: the fixed specialist forms a sharp V about its trained value, while the randomised baseline and the distilled student form similar shallow bowls.

The condition cube reproduces this at population scale (Table 7, buoy column). The baseline holds 2.85 cm on the combined all-stressors condition against 3.3–4.0 cm for the four students, beating every one, and beats Arm A on heavy buoyancy and on payload. The students retain narrow advantages: Arm D leads the isolated buoyancy and payload axes, Arm B leads centre-of-gravity offset, and the students are two to three times tighter at nominal hover. Attitude reverses the ordering outright, the baseline holding 0.44–0.48° on the isolated axes against 0.05–0.25° for the students. Only Arm D, the position leader, is worse than the baseline in attitude because of the high-gain trade its delay training produces. The comparison is therefore a Pareto trade between channels, not a clean win for either side.

Two further tests confirmed the pattern rather than softening it. On held-out vehicles (Table 6) the baseline holds 0.91 cm with 100% success on *all four* unseen banks, better than every student except Arm D. On the mid-episode buoyancy-step scenario, designed as the cleanest demonstration of online adaptation, it tracked the step about as well as the student. A memoryless policy trained across buoyancies need not infer buoyancy to reject the resulting error; it reacts to it.

One asymmetry limited how far these results could be read: this baseline was randomised over *one* identity axis on *one* body, whereas the students were distilled from populations that also varied geometry, centre-of-gravity offset, start attitude and actuation lag, so their remaining advantages could not be attributed to the architecture at all. Thus, three further baselines were trained, one per remaining arm, each copying its arm's randomisation set exactly, and every policy re-run through the identical condition cube. Four results follow, and they do not point the same way.

First, matching the axis closes most of the gap it was supposed to explain. Under a centre-of-gravity offset the buoyancy-only baseline held attitude to only 3.10° against Arm B's 1.05°; the axis-matched baseline reaches **1.20°**, closing roughly 90% of that gap. The advantage withdrawn above was therefore overwhelmingly a property of the training distribution, with a small residual that does favour the student.

Second, the centre-of-gravity condition is a trade between channels, not a ranking. Reading Table 7 across that row: the buoyancy-only baseline holds position and lets attitude drift; the matched baseline does the reverse; Arm B holds *both*. Only the recurrent student manages both channels at once.

Third, adding an axis costs a memoryless policy globally, and a recurrent one almost nothing. The matched baselines carry a high error that barely changes across conditions (Figure 18): DR_cog reads the same settled position error at nominal hover as under centre-of-gravity offset and under an applied moment. Its own hover precision, not

the stressor, dominates every cell. Adding the offset axis moved the baseline’s nominal hover from 0.58 to 1.60 cm; the same axis moved Arm A to Arm B, 0.28 to 0.31 cm. The floor is not an artefact of stopping early since retraining DR_cog for the full budget with early stopping reduced moved settled position only from 1.83 to 1.60 cm while making attitude *worse*. Thus, so the policy moves along a frontier rather than toward the origin. It is not a lucky seed neither: three seeds each give 0.49–0.96 cm for the buoyancy-only baseline against 1.58–1.90 cm for the offset-matched one, ranges that do not overlap.

Fourth, and cutting the other way, the baselines are the more reliable policies at the edge of the distribution. On the held-out banks all four hold 100% success everywhere, including the 2–5 kg boundary where Arm B falls to 53%, Arm C to 67% and Arm A to 71% (Figure 20). Only Arm D matches them.

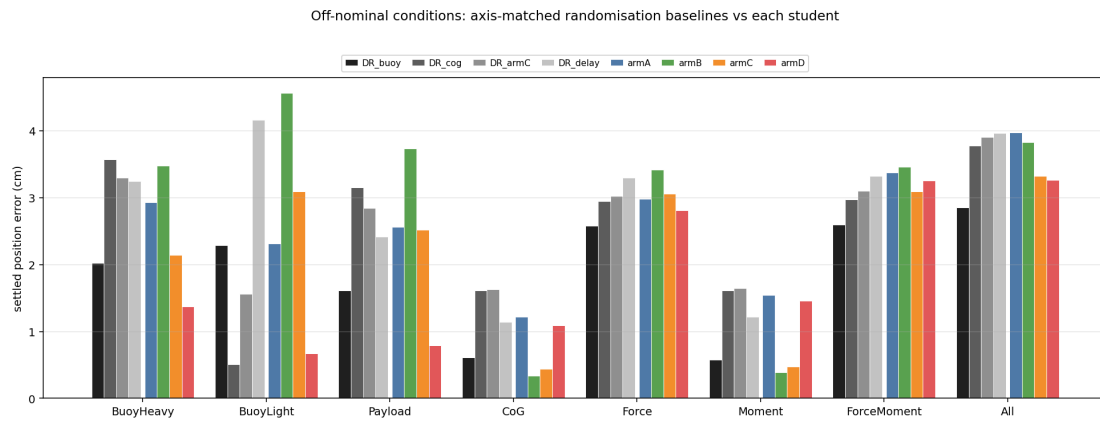


Figure 18: Off-nominal conditions with every baseline matched to its arm’s randomisation axes (greyscale) against the four distilled students (colour). Position error; each grey bar pairs with the coloured bar in the same position.

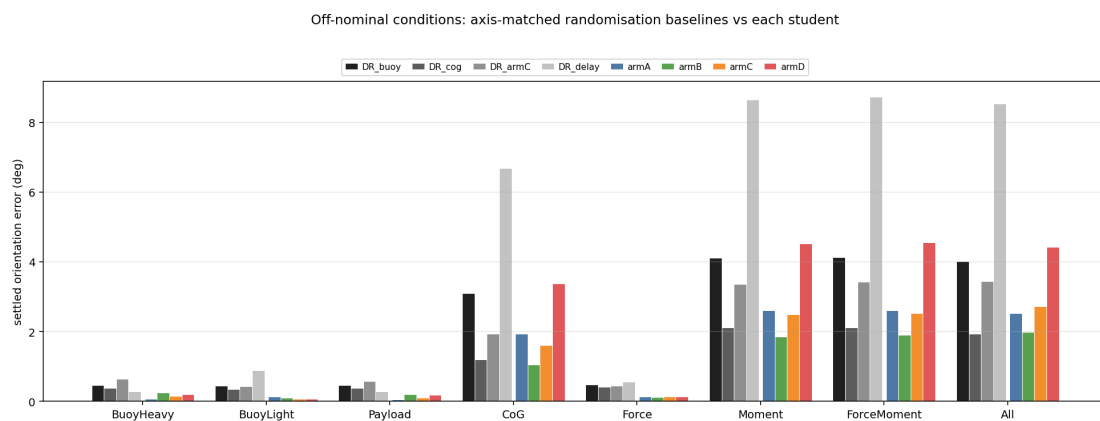


Figure 19: The same comparison in orientation. DR_delay is the outlier: fine at nominal, but it loses attitude wherever a *sustained torque* is applied, because delay training produces a high-gain law and that arm never trained against a persistent moment.

Table 7: Axis-matched baselines against the students: settled error, mean over seven vehicles. Each baseline is paired with the arm directly to its right of the divider (DR_buoy–Arm A, DR_cog–Arm B, DR_armC–Arm C, DR_delay–Arm D). Best in each row in bold.

Condition	randomisation baselines				distilled students			
	buoy	cog	armC	delay	A	B	C	D
<i>settled position error (cm)</i>								
Hover (nominal)	0.58	1.60	1.54	1.09	0.28	0.31	0.39	0.54
Buoyancy heavy	2.03	3.57	3.30	3.25	2.93	3.48	2.14	1.38
Buoyancy light	2.29	0.51	1.56	4.17	2.32	4.57	3.09	0.67
Payload	1.61	3.16	2.84	2.42	2.56	3.74	2.52	0.79
CoG offset	0.61	1.62	1.63	1.14	1.22	0.33	0.44	1.09
Applied moment	0.58	1.61	1.65	1.22	1.54	0.39	0.48	1.46
All combined	2.85	3.78	3.91	3.97	3.97	3.83	3.32	3.26
<i>settled orientation error (deg)</i>								
Hover (nominal)	0.46	0.38	0.37	0.52	0.09	0.08	0.08	0.11
CoG offset	3.10	1.20	1.94	6.69	1.94	1.05	1.61	3.38
Applied moment	4.11	2.11	3.36	8.65	2.60	1.85	2.49	4.53
All combined	4.01	1.94	3.44	8.54	2.52	1.98	2.73	4.42

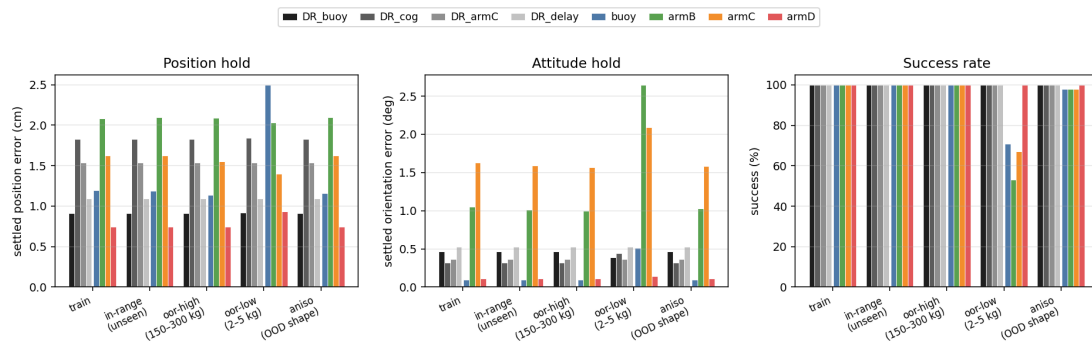


Figure 20: Held-out generalisation, matched baselines against students. Every baseline holds full success on all five banks; three of the four students degrade at the light-mass boundary.

Taken together the matched comparison splits by axis rather than resolving to a verdict: on the condition cube three of the four matched pairs favour the student, on held-out generalisation three of the four favour the baseline, nominal precision favours the students throughout, and combined-condition attitude is won outright by DR_cog. The pipeline’s margin is therefore narrower than assumed, differently signed on different axes, and measured rather than inferred.

4.13 Problems Encountered in Execution

There were three main execution problems. First, the GPU training engine and the double-precision reference engine diverge numerically for high torque-to-inertia vehicles

at the training timestep: a policy-free open-loop test showed the reference integrator diverging to order 10^{36} within one or two steps above a torque-to-inertia ratio of about seven, while the GPU engine stayed stable on the identical input. This is a property of the engines, not the policy, and drove the decision to treat the GPU engine as ground truth.

Second, an early hypothesis that heavy vehicles' policies were chattering was refuted by measurement: the naive sign-flip fraction correlated *negatively* with mass ($r = -0.85$) and the action-rate magnitude was indistinguishable from zero for every teacher tested, the apparent jitter being floating-point noise crossing zero in a visualiser running at mismatched fidelity.

Third, training throughput was unreliable: against a baseline of roughly thirteen minutes per teacher, a sustained slowdown to roughly forty-eight occurred over multi-day runs on the shared machine. ...Swap usage, thread counts, GPU utilisation, thermal state and disk pressure were each measured. The most likely cause was thread and subprocess proliferation per run overwhelming the scheduler at sixteen-way parallelism, though this was not isolated by a controlled test, so it is reported as the leading candidate rather than a confirmed cause.

5 Discussion

This chapter interprets the findings of Chapter 4 against the literature of Chapter 2, taking the four research questions in the order the evidence forced: the mechanism and the identity-parameter condition, which are secure; then transfer; then the baseline comparison, which qualifies the rest.

5.1 Recurrence as Implicit System Identification

The explanation of how this works is the most secure finding of the study (RQ1). A teacher is feedforward: given the same instantaneous error and velocity it must produce the same action, whatever vehicle it is on. The student is recurrent, its hidden state integrates the history of the vehicle's response to its own commands. Until now that mechanism was an argument; the probe makes it evidence. A linear readout of the hidden state predicts buoyancy on held-out vehicles at $R^2 = 0.87$ while both controls collapse to zero, so the encoding is genuine and generalises to vehicles the probe never saw; applied over time, it shows the estimate forming within about three seconds of contact. Online system identification is measured rather than asserted. Reproducing RAPTOR's latent-probe evidence [10] in an independent physical domain, for a different identity parameter, corroborates the account RMA [15] and RAPTOR advance.

The selectivity matters as much as the magnitude: the representation encodes almost nothing but buoyancy, encoding the quantity coupled to the control objective and discarding bulk identity. This presents a cleaner instance of task-relevant abstraction than the aerial result, where the probed parameter is also the bulk one. Its ceiling matters more than it first appears, as Section 5.4 shows: decodability establishes that the information is present, not that the control law exploits it.

5.2 A Condition on Identity-Parameter Handling

RQ2 is answered by the plateau and its resolution, and the answer generalises beyond this project. The randomisation literature treats randomised quantities as nuisance noise resampled every episode [13, 45], correctly for the mass and gravity jitter used here, since the teachers were themselves trained under it. Buoyancy is different in kind: drawn once per teacher and fixed for its lifetime, it is an identity parameter, like which physical body the teacher is. During distillation it must be held at the value the imitated teacher was trained under; resampling evaluates that teacher on physics it never saw, so the target is not merely noisier but systematically wrong. That is why three independent student-side interventions all converged to the identical plateau, nothing on the student side being able to repair targets inconsistent upstream of it. The condition generalises

to any teacher–student pipeline in which teachers are trained under a fixed per-teacher constant; the aerial literature does not confront it because quadrotor control has no buoyancy analogue. It is also the first instance of an epistemic failure mode that recurs, as Section 5.5 develops.

5.3 Where Transfer Held and Where It Failed

RQ3 divides cleanly. Geometry (scale *and* shape) transfers essentially for free: cross-transfer rows are flat, and every policy holds training-level precision at twice the maximum training mass and on 5:1 aspect-ratio hulls. The mechanism is identifiable: because the action is normalised to each vehicle’s own authority and that authority scales with mass, the closed-loop response to a normalised command is approximately mass-invariant by construction. The corollary is that the thirty-fold mass range, which sounds like the hardest test on paper, turns out not to be where the difficulty lies; the only limit found is the light-mass boundary.

Attitude transfer to Stonefish is where the naive expectation of uniform transfer broke, and the two causes must be kept separate. The gross roll oscillation was a ROS2 message-timing artefact, not a control finding. The second is physical: the BlueROV2-Heavy is bottom-heavy, producing a pendulum restoring moment and roll–sway coupling that the idealised centred-mass training vehicles never exhibited. The student met rotational dynamics absent from its training distribution, which strengthens the methodological contribution: randomisation for cross-vehicle transfer must cover the target’s *rotational* physics, not only its mass properties. Each axis must also be bounded by the authority available to counteract it (same lesson as the buoyancy limit).

5.4 What the Pipeline Is Actually Worth

RQ4 is where the study revises itself, and it must be stated plainly. A single feedforward policy trained on one vehicle with buoyancy randomised per episode (ten minutes vs days) matches or beats the distilled students on the combined stressor condition, on single-axis buoyancy, on mid-episode buoyancy change, and on unseen vehicles. The premise the project was built on, that randomising buoyancy is insufficient and inference required, is not supported.

What matters here is not that the premise failed, but why. A memoryless policy randomised over buoyancy cannot know its buoyancy but does not need to: the error the mis-trim produces is itself observable and a well-trained reactive law nulls it. Inferring the parameter and reacting to its consequences are, here, near-substitutable, because buoyancy manifests as a low-frequency force bias fully revealed by the tracking error it causes within a fraction of a second. This is exactly the condition under which

the theoretical argument for memory [15] carries little practical weight: recurrence pays when optimal actions across the distribution genuinely diverge, not merely when the parameter is unobserved. This also explains the connection between the probe and baseline results: the student demonstrably *does* infer buoyancy, but that inference simply is not what the control performance depends on.

What survives is genuine where both sides met the condition: the students are two to three times tighter at nominal hover and four to six times more precise in attitude. One apparent advantage was withdrawn on measurement rather than argument: matching the centre-of-gravity axis closes roughly 90% of Arm B’s attitude lead, leaving a small residual that does favour the student but showing the bulk of what looked architectural to be a property of the training distribution.

What the matched comparison substitutes for that withdrawn claim concerns capacity rather than inference. A single feedforward policy pays for each additional randomisation axis *globally*: giving the baseline the centre-of-gravity axis nearly tripled its nominal hover error, and its error thereafter is condition-invariant (the same at hover as under the offset it was trained for) where the same axis cost the pipeline almost nothing. Retraining three times as long left that floor intact, so it is a frontier rather than a budget artefact. The mechanism is intelligible: a memoryless policy must commit to one control law least bad across everything it was randomised over, whereas a recurrent policy conditions on an inferred context and absorbs axes into the inference. The value of recurrence is therefore not that it adapts better along any one axis but that it *composes* axes without paying for each.

A countervailing result deserves equal weight. On the held-out banks every matched baseline holds full success everywhere, including the light-mass boundary where three of the four students degrade sharply. The explanation follows from Section 5.1: the students infer the vehicle, and an inference outside its training support produces a confident wrong answer, whereas a policy that never learned a body-dependent mapping has nothing to extrapolate wrongly. Implicit system identification is therefore not free: it buys composition inside the training distribution and costs graceful degradation outside it.

5.5 Reflection on the Study’s Own Method

The central findings of this study share a common pattern. The buoyancy plateau happened because the assumption that randomised quantities are just noise to be trained over was borrowed from the randomisation literature without being tested for this specific parameter. The baseline result happened for the same reason: an assumption inherited from the host group’s earlier work, that buoyancy randomisation doesn’t work, was

carried into the project's design without re-checking it, and shaped two months of work as a result.

Why the earlier in-group attempt did not succeed is not diagnosed here, it being improper to speculate about another researcher's implementation; the methodological point stands independently of the cause. A baseline inherited as folklore is not a baseline, and the cheapest experiment in this entire study would, if run first, have reframed everything that followed.

The pattern recurred a third time, when the first account of the baseline result claimed the pipeline retained advantages in attitude and in fault robustness. It was caught in time to be *fixed* rather than merely disclosed, and running the missing baselines converted the study's largest stated limitation into a measured result that both withdrew a claim and produced a better one. The lesson is that a baseline isolates only what it was matched on: matching the evaluation protocol, which this study did from the start, is not matching the training distribution, which it did not until late. The more useful lesson learned from this is: when the missing control is cheap, run it directly.

6 Conclusion

6.1 Summary of the Research

This dissertation asked whether a single recurrent policy, distilled from a population of per-vehicle reinforcement-learning teachers, can control a diverse bank of 6-DOF underwater vehicles without retuning; by what mechanism it adapts; whether that adaptation generalises; and what the architecture is worth against a far cheaper baseline.

All four questions are now answered. A single recurrent student matched the specialists on nominal physics and clearly exceeded them under isolated off-nominal conditions (RQ1, Section 4.7). The mechanism is implicit system identification: a linear probe recovers a held-out vehicle’s hover command from the hidden state almost perfectly, the estimate forming online within roughly three seconds, while bulk body properties sit at the control level (RQ2, Section 4.8). That result depends on treating each teacher’s buoyancy as a fixed constant during distillation since re-randomising it as nuisance noise corrupted the labels into a plateau no student-side intervention could lift. Geometry then transfers essentially for free, at training-level precision on unseen vehicles and at twice the maximum training mass, while attitude transfer to a bottom-heavy vehicle failed, exposing rotational dynamics the training distribution had omitted and pointing to the same fix required to close the gap (RQ3, Sections 4.9–4.11).

RQ4 revises the study’s own premise. A memoryless policy randomised over one axis, at roughly ten minutes of compute against days, matched or exceeded the distilled students on the combined stressor condition and on unseen vehicles. Rather than leave that comparison unmatched, three further baselines were trained, one per remaining arm, and the result divides by axis: matching the centre-of-gravity axis closed roughly 90% of the attitude gap it was supposed to explain, so most of that apparent architectural advantage was a property of the training distribution rather than of recurrence. What replaced it is better supported: a memoryless policy pays for each added axis *globally*, where the pipeline does not. However, it cuts the other way outside the training range, where the baselines degrade gracefully and three of four students do not (Section 4.12).

6.2 Contributions

The study makes five defensible contributions. First, direct evidence that a recurrent distilled policy adapts by implicit system identification, decoding a held-out vehicle’s hover command from the hidden state almost perfectly while bulk properties sit at the control level and forming that estimate online within about three seconds, reproducing in a non-aerial domain evidence previously given only for aerial thrust-to-weight. Second, it transfers teacher–student distillation to the 6-DOF underwater domain across a

5–150 kg bank, the resulting policies controlling unseen vehicles at training-level precision. Third, it establishes a condition for such distillation to work: per-teacher identity parameters must be held fixed during student rollout, and re-randomising them corrupts the labels at a cost easily misdiagnosed as a capacity or observability problem. Fourth, it derives and validates two analytic controllability bounds: buoyancy against thrust authority and centre-of-gravity offset against torque authority; showing randomisation magnitudes must scale to each vehicle’s own span rather than a flat absolute range.

The fifth the study did not set out to make, and is arguably the most useful. Because the randomisation baselines were matched axis by axis rather than only in evaluation protocol, the comparison isolates what recurrence actually buys: not better adaptation along any single axis, where randomisation matches it, but the ability to *compose* axes without a global precision cost and what it costs, namely graceful degradation outside the training distribution. Such cost-revising results are under-reported in a literature where baselines are frequently weaker than the proposed method in ways that are not isolated.

The third and fifth are the strongest publication candidates, and for the same reason: both are transferable claims about *method* rather than about this vehicle class. The identity-versus-nuisance condition applies to any teacher–student pipeline distilling across a parameter distribution, and no prior work states it. The matched-baseline result supplies the controlled comparison the adaptive-control literature generally omits. That submission is planned rather than hypothetical: the work continues in collaboration with Tunçay, whose single-vehicle pipeline [11] this study extends to a population, with **IEEE OCEANS** (held in Scotland in 2027) as the target venue. The items in Section 6.4 would need closing first, seed replication for the two unreplicated baselines, a second sim-to-sim transfer to a vehicle other than the BlueROV2, and sim-to-real, since these are limitations a reviewer would raise before the central claims could stand unqualified.

6.3 Implications

For operators who retune a controller for each vehicle, payload or ballast change, a learned policy can absorb that variation but which form to choose now depends on the deployment envelope, and the matched comparison gives a usable rule. For a single identity axis on a known hull, randomisation is the correct choice, at three orders of magnitude less compute. Where several identity axes vary together, hull, ballast, payload and actuator condition at once, the population approach earns its cost, because a single randomised policy loses precision on every axis as each is added. Where the vehicle may fall outside the range the policy was built for, randomisation is the safer choice: it degrades gracefully where the adaptive policy fails confidently. The engineering

question is therefore not which method is better but whether the fleet sits inside a known envelope, and how many things vary at once.

For engineers building distillation or privileged-learning pipelines the implication is sharper: audit every quantity randomised during distillation and classify it as a per-episode nuisance the teacher itself trained under, or a per-teacher identity constant. The former should be re-randomised, the latter pinned to each teacher's trained value. Getting this wrong throws no error but it silently caps performance in a way easily and expensively misdiagnosed, as it was here across three failed interventions.

Two implications follow for the sim-to-real, meta-reinforcement-learning and marine-robotics communities. Papers distilling across a parameter distribution should state, for each randomised quantity, whether it was treated as per-episode noise or a per-teacher constant, the two not being interchangeable. And this study demonstrates a policy that provably maintains an accurate internal estimate of an identity parameter without that estimate translating into a control advantage over a memoryless policy randomised across the same parameter. Representation-level evidence of adaptation is therefore not evidence that adaptation is what performance depends on. This is a dissociation worth reporting, given that latent probes have become a standard argument for adaptive architectures.

6.4 Limitations

The limitations set the parameters on the value of the work. First, the baseline comparison rests on a single seed for two of the four baselines. The two carrying the axis-cost result were replicated over three seeds each, but the delay and combined arms were not. Additionally, the students were trained on three seeds only for the distillation stage. The axis-matching objection that would otherwise head this list was answered by experiment rather than left standing, but each matched baseline is one training run on one body, so the per-axis conclusions of Section 4.12 carry the uncertainty of a single sample. Second, held-out generalisation uses freshly sampled banks rather than a withheld-teacher split, demonstrating in-distribution generalisation rather than extrapolation from an excluded region. Third, the centre-of-gravity extension is validated in MJX but has not yet been transferred to Stonefish. Fourth, hydrodynamic drag was not randomised. Fifth, all validation is simulated, with no real-hardware trial. Sixth, the numerical divergence between the two engines for high torque-to-inertia vehicles is documented but unresolved, and finally, a training-throughput slowdown that was never conclusively explained. Each bounds what the study establishes and maps to a recommendation in Chapter 7.

6.5 Closing Remarks

The study set out to show that a population-distilled recurrent policy is necessary for underwater buoyancy adaptation, and it succeeded in building one, explaining how it works and showing that it generalises. It did not show that it was necessary. The baseline it was premised on rejecting turned out, when finally measured under the same conditions, to be a strong controller in its own right. Matching that baseline axis by axis dissolved most of what remained of the architectural claim while replacing it with a narrower one that is better evidenced: recurrence does not adapt better along any single axis, it composes several without paying for each, and it forfeits graceful degradation outside the range it was built for.

Reporting that is more valuable than the result the project was commissioned to produce. It saves the next researcher the same two months, it converts the study's largest limitation into a measurement rather than a caveat, and it stands as a demonstration that the cheapest experiment is usually the one most worth running first and that when you notice the control you are missing, the right response is to run it.

7 Recommendations

7.1 Recommendations for Practice

Each recommendation is traceable to a specific result in Chapter 4. First, and most importantly, establish the cheapest credible baseline before committing to a costly architecture, evaluate it under the same protocol, and match it in randomisation breadth as well. An advantage measured on a condition only one side trained for is a property of the training distribution, not the architecture. Section 4.12 is the warrant: ten minutes of compute produced a controller competitive with a pipeline costing three orders of magnitude more. Second, in any distillation pipeline, hold identity parameters fixed per teacher during student rollout and re-randomise only what the teacher itself trained under. Re-randomising a per-teacher constant roughly halved achievable accuracy here and produced a plateau three unrelated interventions could not lift. Third, size randomisation and test magnitudes as a fraction of each vehicle's own controllability span rather than a flat absolute range, since that span varies a lot across a diverse bank, with buoyancy half-span ranging from 0.29 to 1.35. Sampling the offset from geometry alone put the heaviest vehicles into uncontrollable configurations; bounding it by each vehicle's torque authority fixed this. Fourth, select checkpoints by whole-episode RMS across all saved checkpoints rather than taking the final one, since one teacher's final checkpoint scored 3.5 times worse than its best. Fifth, validate any evaluation tool against the actual training engine before trusting it, an apparent actuator-chattering pathology having proved to be a mismatched-fidelity artefact; and read settled hold rather than action traces when judging a high-gain policy, since the delay-trained arm's saturated trace nearly caused its fault robustness to be missed. Sixth, ensure the metric that stops training can see the quantity that will be reported: here the stopping criterion watched whole-episode RMS while the results were settled hover, and two policies differing by 4% on the former differed threefold on the latter which made a genuine capability limit look like an artefact of training budget.

7.2 Recommendations for Further Research

The highest-priority recommendation follows from the one comparison the matched baselines could not repair. The actuator-fault sweep never included a randomisation baseline, so the delay-trained arm's robustness to unseen thruster faults compares two students against a fixed specialist and cannot be attributed to the architecture. Running the matched baselines through that sweep costs an afternoon and would either establish the result or remove it; just like the equivalent audit on the condition cube overturned one claim and produced a better one in its place.

Second, the per-axis findings of Section 4.12 each rest on a single run on a single body. Repeating the four matched baselines across several seeds and bodies would make them statistically defensible, at a cost of hours; the global precision cost of adding axes is the study's strongest architectural claim and currently its least replicated.

Third, and most interesting scientifically, the light-mass failure mode deserves direct study. The students degrade there while every baseline does not, suggesting an identity estimate outside its training support produces a confident wrong answer rather than a cautious one. Whether such a controller can be made to *detect* that it has left its training distribution, and fall back to a conservative law, would matter to any deployed adaptive system.

Fourth, the recurrence–performance relationship of Section 5.4 deserves direct study on more than one parameter. This work showed a policy that definitely encodes an identity parameter yet gains no control advantage from doing so along that axis alone, while gaining a clear one once several axes vary together. Characterising *which* parameters reward inference individually, presumably those whose optimal responses diverge sharply across the distribution, or whose effects are not promptly revealed in the tracking error, and how that interacts with the number of axes in play, would give a principled criterion for when memory is worth its cost.

Fifth, replace the freshly sampled held-out banks with a strict withheld-teacher split, which measures extrapolation rather than in-distribution generalisation. But also which the light-mass result makes considerably more interesting than it was. Sixth, run the centre-of-gravity policy through the same sim-to-sim attitude transfer test as the others. Seventh, add hydrodynamic drag to the randomisation. Eighth, add early stopping to distillation but on a settled-window metric, not whole-episode RMS, whose insensitivity to steady-state hover is what made the baselines' apparent convergence misleading in the first place. Finally, progress to real-hardware trials, which the identity-parameter, fault-robustness and baseline findings most need to be established beyond simulation.

A caution attaches to designing any of these comparisons: matching the evaluation protocol is not sufficient. Training distributions must be matched axis by axis, and any axis given to one side only excluded from architectural claims. The natural enforcement is to fix the randomisation specification first and derive both the baseline and the teacher population from it, so the two cannot silently diverge.

7.3 Reflective Summary

This project developed skills across reinforcement learning, GPU-parallel simulation, marine vehicle dynamics, ROS2 integration and large-scale experiment management,

but its most durable lesson concerned engineering judgement rather than technique, and is drawn out in Section 5.5. On sustainability: the two architectures differ by roughly three orders of magnitude of energy, so establishing where the cheaper suffices is itself an environmental result. On ethics and integrity, reporting a result that undercuts the project's own premise was the only defensible course, and produced a stronger dissertation than concealing it would have. On equality, diversity and inclusion, the relevant dimension is access: a population-scale method reproducible only on a multi-GPU cluster is unavailable to researchers without one, so the cheaper baseline proving competitive is an inclusion result as well as an efficiency one and the analysis code, banks and protocols have been kept runnable on a single consumer GPU so the findings can be checked without privileged resources.

References

- [1] R. B. Wynn *et al.*, “Autonomous underwater vehicles (AUVs): Their past, present and future contributions to the advancement of marine geoscience,” *Marine Geology*, vol. 352, pp. 451–468, 2014. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0025322714000747>
- [2] A. Shukla and H. Karki, “Application of robotics in offshore oil and gas industry— a review part II,” *Robotics and Autonomous Systems*, vol. 75, pp. 508–524, 2016. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0921889015002018>
- [3] M. Jacobi, “Autonomous inspection of underwater structures,” *Robotics and Autonomous Systems*, vol. 67, pp. 80–86, 2015. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0921889014002267>
- [4] J. G. Bellingham and K. Rajan, “Robotics in remote and hostile environments,” *Science*, vol. 318, no. 5853, pp. 1098–1102, 2007. [Online]. Available: <https://www.science.org/doi/abs/10.1126/science.1146230>
- [5] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed. Wiley, 2021. [Online]. Available: <https://onlinelibrary.wiley.com/doi/book/10.1002/9781119575016>
- [6] —, *Hydrostatics*. John Wiley & Sons, Ltd, 2021, ch. 4, pp. 71–104. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781119575016.ch4>
- [7] G. Antonelli, *Underwater Robots*, ser. Springer Tracts in Advanced Robotics. Cham: Springer International Publishing, 2014, vol. 96. [Online]. Available: <https://link.springer.com/10.1007/978-3-319-02877-4>
- [8] K. Zakka *et al.*, “MuJoCo Playground,” 2025. [Online]. Available: <https://arxiv.org/abs/2502.08844>
- [9] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033. [Online]. Available: <https://ieeexplore.ieee.org/document/6386109/>
- [10] J. Eschmann, D. Albani, and G. Loianno, “RAPTOR: A foundation policy for quadrotor control,” *Science Robotics*, vol. 11, no. 114, p. eaec1481, 2026. [Online]. Available: <https://www.science.org/doi/abs/10.1126/scirobotics.aec1481>

- [11] S. Tunçay, A. Andres, and I. Carlucho, “Fast Policy Learning for 6-DOF Position Control of Underwater Vehicles,” 2025. [Online]. Available: <https://arxiv.org/abs/2512.13359>
- [12] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., ser. Adaptive Computation and Machine Learning. Cambridge, Massachusetts: The MIT Press, 2018.
- [13] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain randomization for transferring deep neural networks from simulation to the real world,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30. [Online]. Available: <https://doi.org/10.48550/arXiv.1703.06907>
- [14] S. Ross, G. J. Gordon, and J. A. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS)*, vol. 15, 2011, pp. 627–635. [Online]. Available: <https://doi.org/10.48550/arXiv.1011.0686>
- [15] A. Kumar, Z. Fu, D. Pathak, and J. Malik, “RMA: Rapid motor adaptation for legged robots,” in *Robotics: Science and Systems (RSS)*, 2021. [Online]. Available: <https://arxiv.org/abs/2107.04034>
- [16] M. Grimaldi *et al.*, “Stonefish: Supporting machine learning research in marine robotics,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.11887>
- [17] T. Chaffre *et al.*, “Sim-to-real transfer of adaptive control parameters for AUV stabilisation under current disturbance,” *The International Journal of Robotics Research*, vol. 44, no. 3, pp. 407–430, 2025. [Online]. Available: <https://doi.org/10.1177/02783649241272115>
- [18] L. Deng and J. Tao, “Thrust allocation control of an underwater vehicle with a redundant thruster configuration,” *Mathematics*, vol. 13, no. 11, p. 1766, 2025. [Online]. Available: <https://doi.org/10.3390/math13111766>
- [19] T. Prestero, “Verification of a six-degree of freedom simulation model for the REMUS autonomous underwater vehicle,” Master’s thesis, Massachusetts Institute of Technology and Woods Hole Oceanographic Institution, 2001. [Online]. Available: <http://hdl.handle.net/1721.1/65068>

- [20] T. I. Fossen and T. A. Johansen, "A survey of control allocation methods for ships and underwater vehicles," in *14th Mediterranean Conference on Control and Automation (MED)*, 2006. [Online]. Available: <https://doi.org/10.1109/MED.2006.328749>
- [21] A. M. Eissa, S. A. Mohamed, M. I. Awad, H. E. Abd El Munim, and D. Emad, "Trajectory tracking control of autonomous underwater vehicles: a review from classical methods to AI-based approaches," *International Journal of Dynamics and Control*, 2025. [Online]. Available: <https://doi.org/10.1007/s40435-025-01963-5>
- [22] L. He, M. Xie, and Y. Zhang, "A review of path following, trajectory tracking, and formation control for autonomous underwater vehicles," *Drones*, vol. 9, no. 4, p. 286, 2025. [Online]. Available: <https://doi.org/10.3390/drones9040286>
- [23] R. Makam, P. Mane, S. Sundaram, and P. B. Sujit, "A comprehensive study on modelling and control of autonomous underwater vehicle," 2023. [Online]. Available: <https://arxiv.org/abs/2312.02690>
- [24] M. Hedayati Khodayari and S. Balochian, "Modeling and control of autonomous underwater vehicle (AUV) in heading and depth attitude via self-adaptive fuzzy PID controller," *Journal of Marine Science and Technology*, vol. 20, no. 3, pp. 559–578, 2015. [Online]. Available: <https://doi.org/10.1007/s00773-015-0312-7>
- [25] D. R. Yoerger and J.-J. E. Slotine, "Robust trajectory control of underwater vehicles," *IEEE Journal of Oceanic Engineering*, vol. 10, no. 4, pp. 462–470, 1985. [Online]. Available: <https://doi.org/10.1109/JOE.1985.1145131>
- [26] A. J. Healey and D. Lienard, "Multivariable sliding mode control for autonomous diving and steering of unmanned underwater vehicles," *IEEE Journal of Oceanic Engineering*, vol. 18, no. 3, pp. 327–339, 1993. [Online]. Available: <https://doi.org/10.1109/JOE.1993.236372>
- [27] J. Xu, M. Wang, and L. Qiao, "Dynamical sliding mode control for the trajectory tracking of underactuated unmanned underwater vehicles," *Ocean Engineering*, vol. 105, pp. 54–63, 2015. [Online]. Available: <https://doi.org/10.1016/j.oceaneng.2015.06.022>
- [28] J. Zhou, X. Zhao, Z. Feng, and D. Wu, "Trajectory tracking sliding mode control for underactuated autonomous underwater vehicles with time delays," *International Journal of Advanced Robotic Systems*, vol. 17, no. 3, 2020. [Online]. Available: <https://doi.org/10.1177/1729881420916276>

- [29] G. Antonelli, S. Chiaverini, N. Sarkar, and M. West, “Adaptive control of an autonomous underwater vehicle: experimental results on ODIN,” *IEEE Transactions on Control Systems Technology*, vol. 9, no. 5, pp. 756–765, 2001. [Online]. Available: <https://doi.org/10.1109/87.944470>
- [30] D. A. Smallwood and L. L. Whitcomb, “Model-based dynamic positioning of underwater robotic vehicles: theory and experiment,” *IEEE Journal of Oceanic Engineering*, vol. 29, no. 1, pp. 169–186, 2004. [Online]. Available: <https://doi.org/10.1109/JOE.2003.823312>
- [31] T. Xie, Y. Li, Y. Jiang, L. An, and H. Wu, “Backstepping active disturbance rejection control for trajectory tracking of underactuated autonomous underwater vehicles with position error constraint,” *International Journal of Advanced Robotic Systems*, vol. 17, no. 2, 2020. [Online]. Available: <https://doi.org/10.1177/1729881420909633>
- [32] H. Chen, G. Tang, S. Wang, W. Guo, and H. Huang, “Adaptive fixed-time backstepping control for three-dimensional trajectory tracking of underactuated autonomous underwater vehicles,” *Ocean Engineering*, vol. 275, p. 114109, 2023. [Online]. Available: <https://doi.org/10.1016/j.oceaneng.2023.114109>
- [33] L. Dai, D. Zhang, S. Wang, G. Nie, X. Zhu, and X. Wang, “Trajectory tracking control of an autonomous underwater vehicle under disturbance and model uncertainty,” *Journal of Marine Science and Engineering*, vol. 13, no. 11, p. 2210, 2025. [Online]. Available: <https://doi.org/10.3390/jmse13112210>
- [34] I. Carlucho, M. De Paula, S. Wang, Y. Petillot, and G. G. Acosta, “Adaptive low-level control of autonomous underwater vehicles using deep reinforcement learning,” *Robotics and Autonomous Systems*, vol. 107, pp. 71–86, 2018. [Online]. Available: <https://doi.org/10.1016/j.robot.2018.05.016>
- [35] V. Mnih *et al.*, “Human-level control through deep reinforcement learning,” *Nature*, vol. 518, no. 7540, pp. 529–533, 2015. [Online]. Available: <https://www.nature.com/articles/nature14236>
- [36] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal Policy Optimization Algorithms,” 2017. [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [37] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” *arXiv*, 2018. [Online]. Available: <https://www.semanticscholar.org/paper/811df72e210e20de99719539505da54762a11c6d>

- [38] S. Fujimoto, H. Van Hoof, and D. Meger, “Addressing function approximation error in actor-critic methods,” 2018. [Online]. Available: <https://arxiv.org/abs/1802.09477>
- [39] Z. Tian, S. Deng, Z. Wu, and C. Zhou, “Reinforcement learning applications in unmanned underwater vehicles: A review,” *Journal of Marine Science and Application*, 2025. [Online]. Available: <https://doi.org/10.1007/s11804-025-00723-3>
- [40] Y. Hsu, H. Wu, K. You, and S. Song, “A selected review on reinforcement learning based control for autonomous underwater vehicles,” 2019. [Online]. Available: <https://arxiv.org/abs/1911.11991>
- [41] Z. Wang, X. Xiang, Y. Duan, and S. Yang, “Adversarial deep reinforcement learning based robust depth tracking control for underactuated autonomous underwater vehicle,” *Engineering Applications of Artificial Intelligence*, vol. 130, p. 107728, 2024. [Online]. Available: <https://doi.org/10.1016/j.engappai.2023.107728>
- [42] G. Boré, V. Sufán, S. Rodríguez-Martínez, and G. Troni, “Toward 6-DOF autonomous underwater vehicle energy-aware position control based on deep reinforcement learning: Preliminary results,” 2025. [Online]. Available: <https://arxiv.org/abs/2502.17742>
- [43] S. Narvekar, B. Peng, M. Leonetti, J. Sinapov, M. E. Taylor, and P. Stone, “Curriculum learning for reinforcement learning domains: a framework and survey,” *Journal of Machine Learning Research*, vol. 21, no. 181, pp. 1–50, 2020. [Online]. Available: <https://doi.org/10.48550/arXiv.2003.04960>
- [44] C. Lu, J. Kuba, A. Letcher, L. Metz, C. Schroeder de Witt, and J. Foerster, “Discovered policy optimisation,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 455–16 468, 2022. [Online]. Available: <https://doi.org/10.52202/068431-1197>
- [45] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, “Sim-to-real transfer of robotic control with dynamics randomization,” in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 1–8. [Online]. Available: <https://doi.org/10.1109/ICRA.2018.8460528>
- [46] W. Lu, K. Cheng, and M. Hu, “Reinforcement learning for autonomous underwater vehicles via data-informed domain randomization,” *Applied Sciences*, vol. 13, no. 3, p. 1723, 2023. [Online]. Available: <https://doi.org/10.3390/app13031723>

- [47] G. Xie *et al.*, “EasyUUV: An LLM-enhanced universal and lightweight sim-to-real reinforcement learning framework for UUV attitude control,” 2025. [Online]. Available: <https://arxiv.org/abs/2510.22126>
- [48] OpenAI *et al.*, “Solving Rubik’s cube with a robot hand,” 2019. [Online]. Available: <https://doi.org/10.48550/arXiv.1910.07113>
- [49] F. Muratore, F. Ramos, G. Turk, W. Yu, M. Gienger, and J. Peters, “Robot learning from randomized simulations: A review,” *Frontiers in Robotics and AI*, vol. 9, p. 799893, 2022. [Online]. Available: <https://doi.org/10.3389/frobt.2022.799893>
- [50] Y. Duan, J. Schulman, X. Chen, P. L. Bartlett, I. Sutskever, and P. Abbeel, “RL²: Fast reinforcement learning via slow reinforcement learning,” 2016. [Online]. Available: <https://arxiv.org/abs/1611.02779>
- [51] V. Makoviychuk *et al.*, “Isaac Gym: High performance GPU-based physics simulation for robot learning,” 2021. [Online]. Available: <https://arxiv.org/abs/2108.10470>
- [52] C. D. Freeman, E. Frey, A. Raichuk, S. Girgin, I. Mordatch, and O. Bachem, “Brax – a differentiable physics engine for large scale rigid body simulation,” 2021. [Online]. Available: <https://arxiv.org/abs/2106.13281>
- [53] J. Bradbury *et al.*, “JAX: composable transformations of Python+NumPy programs,” 2018. [Online]. Available: <https://github.com/google/jax>
- [54] M. Hessel and D. Budden, “Using JAX to accelerate our research,” 2020. [Online]. Available: <https://deepmind.google/blog/using-jax-to-accelerate-our-research/>
- [55] S. Aldhaheri, Y. Hu, Y. Xie, P. Wu, D. Kanoulas, and Y. Liu, “Underwater robotic simulators review for autonomous system development,” 2025. [Online]. Available: <https://arxiv.org/abs/2504.06245>
- [56] J. Heek *et al.*, “Flax: A neural network library and ecosystem for JAX,” 2020. [Online]. Available: <https://github.com/google/flax>
- [57] DeepMind *et al.*, “The DeepMind JAX ecosystem,” 2020. [Online]. Available: <https://github.com/google-deepmind/optax>

Appendices

Appendix 1: Training Hyperparameters and Reward Weights

The values below are those actually used for every result in Chapter 4, read from the training scripts and from the `config.json` written into each run directory. They are recorded here rather than in Chapter 3 because they are reference detail: no argument in the main text turns on any individual value.

Table 8: PPO teacher training. One teacher per hardware configuration.

Parameter	Value	Note
Parallel environments	4,096	batch size justified in §4.3
Episode length	512 steps	10.24 s at 50 Hz
PPO updates per teacher (cap)	301	early stopping usually fires first
Actor architecture	256×256 , tanh	diagonal Gaussian head
Critic architecture	256×256 , tanh	
Weight initialisation	orthogonal	scale 0.01 on the policy mean head
Policy log-std	state-independent, learned	initialised to 0, i.e. $\sigma = 1$
Actor learning rate	3×10^{-4}	Adam
Critic learning rate	1×10^{-4}	Adam
Gradient clipping	global norm 0.5	applied to both networks
PPO epochs per update	4	passes over each rollout
Minibatch size	$4 \times \text{num_envs}$	16,384 transitions
Clipping ϵ	0.2	
Discount γ	0.99	
GAE λ	0.95	
Value-loss coefficient	0.5	
Entropy coefficient	0.01	
Policy std. bounds	[0.01, 1.0]	clipped for numerical stability
<i>Early stopping</i> (evaluated every 5 updates)		
Minimum updates before check	60	
Patience, each metric	15 updates	position and orientation
Patience, summed	40 updates	
Relative improvement counted	3 %	
Sanity floor	0.05	per metric; will not stop above it

Table 9: Student distillation. Identical across all four arms; only the observation dimension differs, being 18 rather than 12 for the actuation-delay arms.

Parameter	Value	Note
Parallel environments	128	per teacher, per epoch
Episode length	512 steps	full BPTT window
Epochs	100	200 for the nobuoy reference
Warm-up epochs	10	teacher-driven rollouts
GRU hidden dimension	16	64 tested to rule out a capacity bottleneck
Latent dimension	16	
Learning rate	3×10^{-4}	Adam
Gradient clipping	global norm 1.0	
Loss	MSE on teacher mean	backpropagation through time

Table 10: Reward weights, for Equation 3. The whole sum is multiplied by a return scale of 5×10^{-4} . Position uses per-axis weights and an additional squared-norm term.

Term	Weight	Purpose
Position error, per axis x, y, z	3.0 each	translational tracking
Position squared norm	10.0	sharpens the approach to zero
Axis-angle orientation error	100.0	attitude tracking
Command magnitude	5.0	effort penalty
Body velocity	5.0	damping
Deviation from moving average	0.01	smoothness
Change from previous command	1.0	rate penalty

Table 11: Domain-randomisation constants. The first two are per-episode nuisance jitter; the remainder are the per-teacher identity axes of Section 3.7.

Quantity	Value	Category
Mass jitter	$\pm 0.1\%$	per-episode nuisance
Gravity jitter	$\pm 0.01 \text{ m s}^{-2}$	per-episode nuisance
Buoyancy draw	70 % of own span	per-teacher identity
CoG offset cap	50 % of own torque span	per-teacher identity
CoG offset, lateral	20 % of semi-axis	geometric cap, symmetric
CoG offset, vertical	50 % of semi-axis	geometric cap, one-sided
Start attitude	$\pm \pi/2$ per axis	arms B and C
Actuation lag τ	50 ms	arms C and D, $\alpha \approx 0.29$
Hydrodynamic drag	not randomised	stated limitation, §6.4

Appendix 2: Reproducibility Record

Every reported number traces to a named run directory containing the checkpoints, the per-epoch training log, the hardware manifest and the `config.json` from which the tables in Appendix 4 were read.

Table 12: Software environment, identical across all machines.

Component	Version	Component
Python	3.12	jax / jaxlib 0.10.0
mujoco / mujoco-mjx	3.8.1	flax 0.10.6
numpy	2.4.4	optax 0.2.4
distrax	0.1.8	tyro 0.9.22

Table 13: Teacher banks. Every bank uses the same 100-body hardware bank with ten buoyancy variants per body, giving 1,000 teachers; nobuoy is the one-teacher-per-body reference predating buoyancy diversity.

Arm	Run directory	CoG	Start	Lag	Variants
reference	merged_n1000	—	identity	—	1
A	merged_buoyancy1000	—	identity	—	10
B	merged_cog1000_corrected	yes	$\pm\pi/2$	—	10
C	merged_armC1000	yes	$\pm\pi/2$	50 ms	10
D	merged_armD1000	—	identity	50 ms	10

Table 14: Distilled students, as recorded in each run’s `config.json`. Runtimes are the distillation stage only and exclude teacher training.

Label	Run directory	Machine	Epochs	Obs.	Runtime
nobuoy	20260710_110054	laptop	200	12	1.7 h
buoy (A)	20260717_102816	osl	100	12	7.4 h
Arm B	20260727_174851	osl	100	12	9.2 h
Arm C	20260729_114355	HWLL0149	100	18	4.4 h
Arm D	20260729_185011	HWLL0149	100	18	4.5 h

Teacher training was distributed across the same three machines as separate processes over disjoint teacher-index ranges, then merged into the single bank directories above. Evaluation outputs are written to `runs/scenario/`, `runs/probe/` and the condition-cube directories, and every figure in Chapter 4 is regenerated directly from those files by the analysis scripts.

Appendix 3: Condition-Cube Test Vehicles

The condition cube of Section 4.7 evaluates every policy on seven vehicles spanning the mass range, one drawn from each mass class of the hardware bank. Their sampled physical parameters are given below; all derive from the mass-scaling laws of Section 3.4.

Table 15: The seven class-representative test vehicles. Semi-axes a , b , c define the hull ellipsoid; thrust is the per-axis linear limit and τ the per-axis torque limit.

Class	Bank id	Mass [kg]	a [m]	b [m]	c [m]	Thrust [N]	τ_{roll} [N m]	τ_{yaw} [N m]
mini	5	5.8	0.220	0.206	0.126	55.0	5.9	6.4
small	19	9.8	0.307	0.220	0.115	104.7	6.8	5.8
medium	35	15.2	0.268	0.266	0.187	102.6	9.7	14.5
worklight	52	28.0	0.403	0.303	0.176	138.9	17.5	22.1
workmed	70	60.9	0.427	0.410	0.297	340.0	36.7	38.7
compact	82	82.8	0.493	0.415	0.281	281.7	95.0	59.8
large	93	126.1	0.720	0.481	0.308	575.3	109.7	91.4

The specialist teacher columns of the cube are the buoyancy variant of each class representative closest to neutral, so that the comparison is not confounded by a teacher carrying a large trained buoyancy bias into a neutral test (Table 16). Three classes required a substituted variant: the nearest-to-neutral draw for those bodies was a poorly converged teacher, and using it would have flattered the students.

Table 16: Specialist teachers used as the cube’s teacher columns.

Class	Teacher id	Buoyancy ratio	Selection
mini	58	0.997	nearest neutral
small	191	1.013	nearest neutral
medium	359	1.069	nearest neutral
worklight	521	0.985	nearest neutral
workmed	700	1.123	substituted, best converged
compact	811	1.001	substituted, best converged
large	934	0.840	substituted, best converged

Appendix 4: Supplementary Video Index

The behaviours reported in Chapter 4 are dynamic, and several are more legible in motion than in a settled-error figure. Short clips isolating each result are listed first, followed by full-length runs for readers who want to see a complete evaluation rather than an excerpt. Each is silent and captioned on screen with the vehicle, the condition applied and the policy shown.

	Title	What it shows	Section
V1	Specialist versus distilled student	Both policies on the <i>same</i> body while buoyancy is swept from 0.35 to 1.78. The specialist, trained at 0.76, degrades from 4 to 15 cm across the sweep; the student holds within a few centimetres throughout.	§4.7
V2	Hardware swap with hidden state preserved	The student hovering, then the vehicle exchanged mid-flight for a much heavier one without resetting the recurrent state; the visible re-convergence is the adaptation latency measured in Figure 12.	§4.8
V3	Buoyancy driven to the controllability limit	Buoyancy swept outward until the analytic bound of Equation 6 is reached, where hold collapses as predicted.	§4.6
V4	Mid-flight payload, zero-shot	A payload added during hover, a condition present in no training distribution; the specialist sinks, the student compensates.	§4.7
V5	Sim-to-sim transfer, BlueROV2 in Stonefish	The delay-free student (Arm A) and then the delay-trained one (Arm D) driven in turn through the identical trajectory in an independent engine; the vertical limit cycle of the former against the steady hover of the latter.	§4.11
V6	Thruster-fault sweep	Station-keeping under the severe <code>asym</code> configuration, an out-of-distribution actuator fault no policy trained for, where Arm D holds 7.6 cm against the specialist's 18.7 and Arm A's 21.7.	§4.11
V7	Randomisation baseline versus student	The control condition for RQ4 alongside a distilled student, buoyancy swept over the same range as V1. Both hold across it, which is the RQ4 result.	§4.12

Extended runs. The clips above isolate one result each. The runs below are complete evaluations, provided so that a reader can see an entire schedule rather than the phases that favour a particular conclusion.

	Title	What it shows	Section
E1	The 17-phase scenario, end to end	The full 102 s stress schedule of Table 3, specialist and student stepping it in lock-step on the same body, with the active phase and conditions captioned throughout. This is the run behind Figure 9, and it includes the moving-target phases where the student is the weaker of the two.	§4.7
E2	Stonefish dual, tilt schedule	Arm C and Arm D on two BlueROV2s in a single Stonefish window, driven through the same large-attitude schedule at the same instant, from level hold through 60° and 120° tilts to a 160° inversion and back.	§4.11
E3	Randomisation baseline against the student, interactive	The RQ4 comparison driven by hand rather than by a script, both policies receiving the identical command each frame while target pose, buoyancy, payload and disturbance are varied live. Full telemetry is shown for each side, including orientation error and action saturation.	§4.12
E4	The randomisation baseline in Stonefish	The baseline transferred to the BlueROV2 in the independent engine. Reported here as a qualitative observation only: §4.11 measures the four students on this transfer, and no corresponding figure was recorded for the baseline.	§4.11
E5	The student in Stonefish, driven by hand	Free-form pose commands issued interactively to the distilled student on the BlueROV2, rather than a scripted schedule. Recorded after the message-timing correction described in §4.11, so the attitude behaviour shown is the control law's and not the pipeline artefact that preceded it.	§4.11

Links.

- V1 — <https://youtu.be/UGJmcUIegGg>
- V2 — <https://youtu.be/iNicQTPlfDs>
- V3 — <https://youtu.be/0z2d5RWMgxU>
- V4 — <https://youtu.be/Tbl0ceWNGrY>
- V5 — <https://youtu.be/btPi0lSPvYA>
- V6 — <https://youtu.be/dMx9MxsK5fM>
- V7 — <https://youtu.be/AzwFrnhTi2Q>
- E1 — <https://youtu.be/DIOe1ay6aWY>
- E2 — <https://youtu.be/g-vFBkhs4zE>
- E3 — <https://youtu.be/EVKFf9hQa78>
- E4 — https://youtu.be/4RkMvup5_Fg
- E5 — <https://youtu.be/CkhysE8fdVM>